

MW1112 Kernfusion-Reaktortechnik

Nuclear Fusion Reactor Engineering

Lecture 12

Assembly and Commissioning

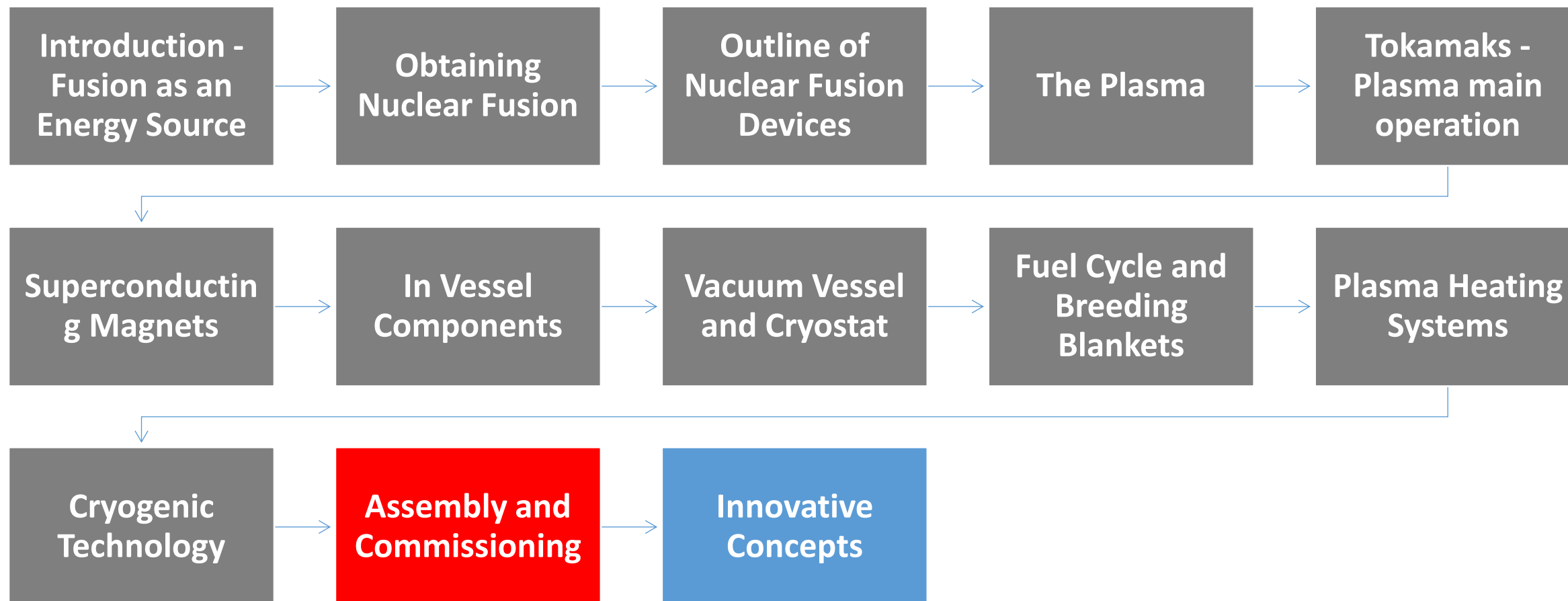
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Assembly and Commissioning

MW1112 Kernfusion-Reaktortechnik – Nuclear Fusion Reactor Engineering

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Fusion Experiment Assembly

MW1112 Kernfusion-Reaktortechnik – Nuclear Fusion Reactor Engineering

Lecture 12 – Assembly and Commissioning

- Transportation to assembly site
- Incoming inspections
- Handling of large components
- Precise positioning and measurement
- Mechanical connections
- Welding and NDE on site
- Electrical connections and insulation

- In most cases, the production site of the components of large experimental devices do not coincides with the assembly site.
- In this case the components must be shipped, either per land, sea or – exceptionally – air, to reach the assembly site.
- This special transports are at the top end of the spectrum of the commercial shipping companies and require careful planning and dedicated tools.
- In particular, for projects of strategic importance, like ITER, the local authorities have widened roads, reinforced bridges and modified intersections in preparation for the exceptional size and weight of some of the Tokamak components.
- For ITER it is expected to have ~250 exceptional convoys to travel, at night, between the harbour area of Marseille and the assembly site in Cadarache.
- The largest ITER convoys have impressive parameters: the heaviest will weigh approximately 800 tonnes (including the 200-tonne, 352-wheel transport vehicle); the tallest will be 10.4 metres high, the longest 33 metres, and the widest 11.5 metres.







The widest ITER transportation
11.5 m for PF6



MAMMOET prototype 342-wheel self-propelled transport vehicle
Twin hydraulic "power packs" (equivalent to four truck tractors)
Plus two conventional truck tractors (Mercedes ACTROS SLT)
Total moving power > 2000 HP



**One of the heaviest ITER transportation
800 t for PF6**





**One of the heaviest ITER transportation
600 t for vacuum vessel sector**





**The longest ITER transportation
35 m for quench tank**





**An example of fluvial transportation from Switzerland
Cryogenic termination box**

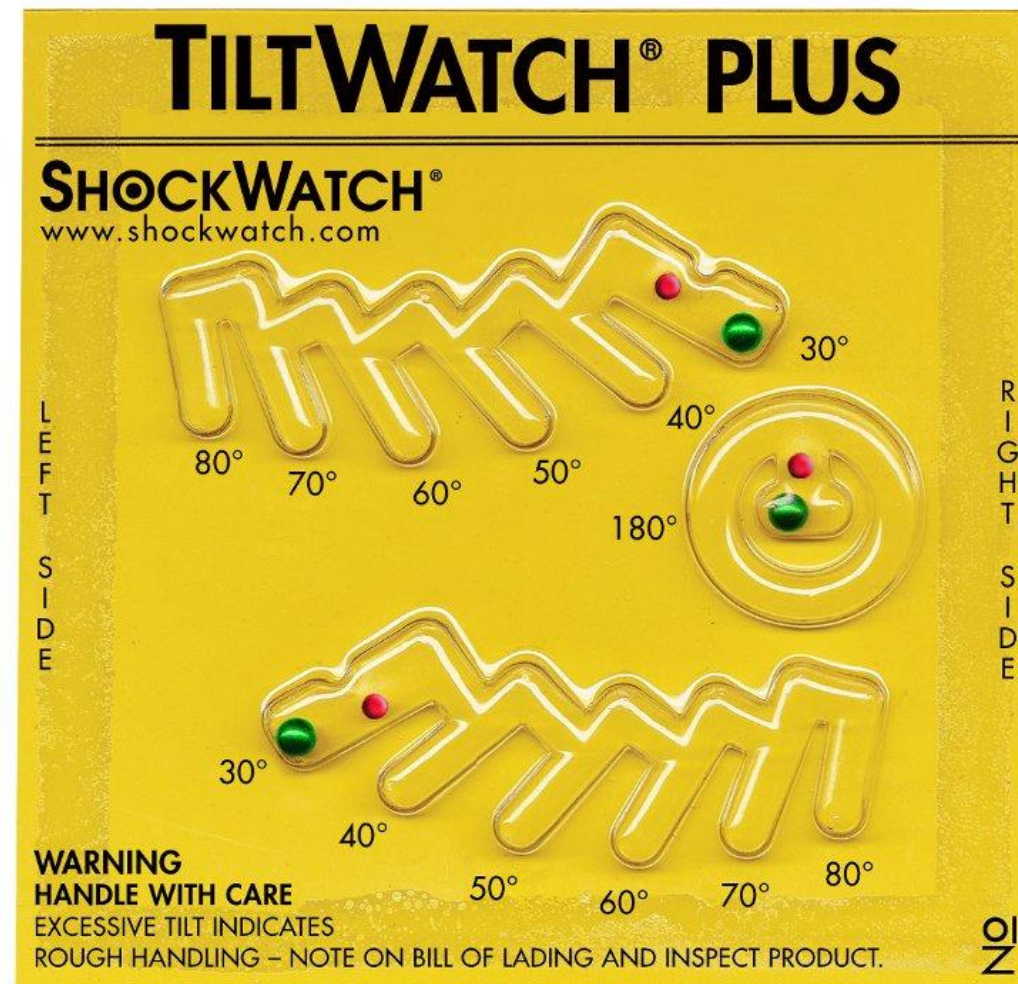




**A (rare) example of air transportation of Tokamak components
JT-60SA Toroidal Field coils (2) → Total weight ~ 80 tons**



- Once the tokamak components arrive on site, and prior to their installation, a comprehensive series of tests is carried out to make sure that no damage is occurred during transportation. For example:
 - Visual test of the integrity of the packaging
 - Checking of the acceleration sensors, temperature loggers
 - Visual test of the component
 - Pressure, flow and leak tests
 - Electrical insulation tests
 - Dimensional test



<https://www.shockwatch.com.au/>

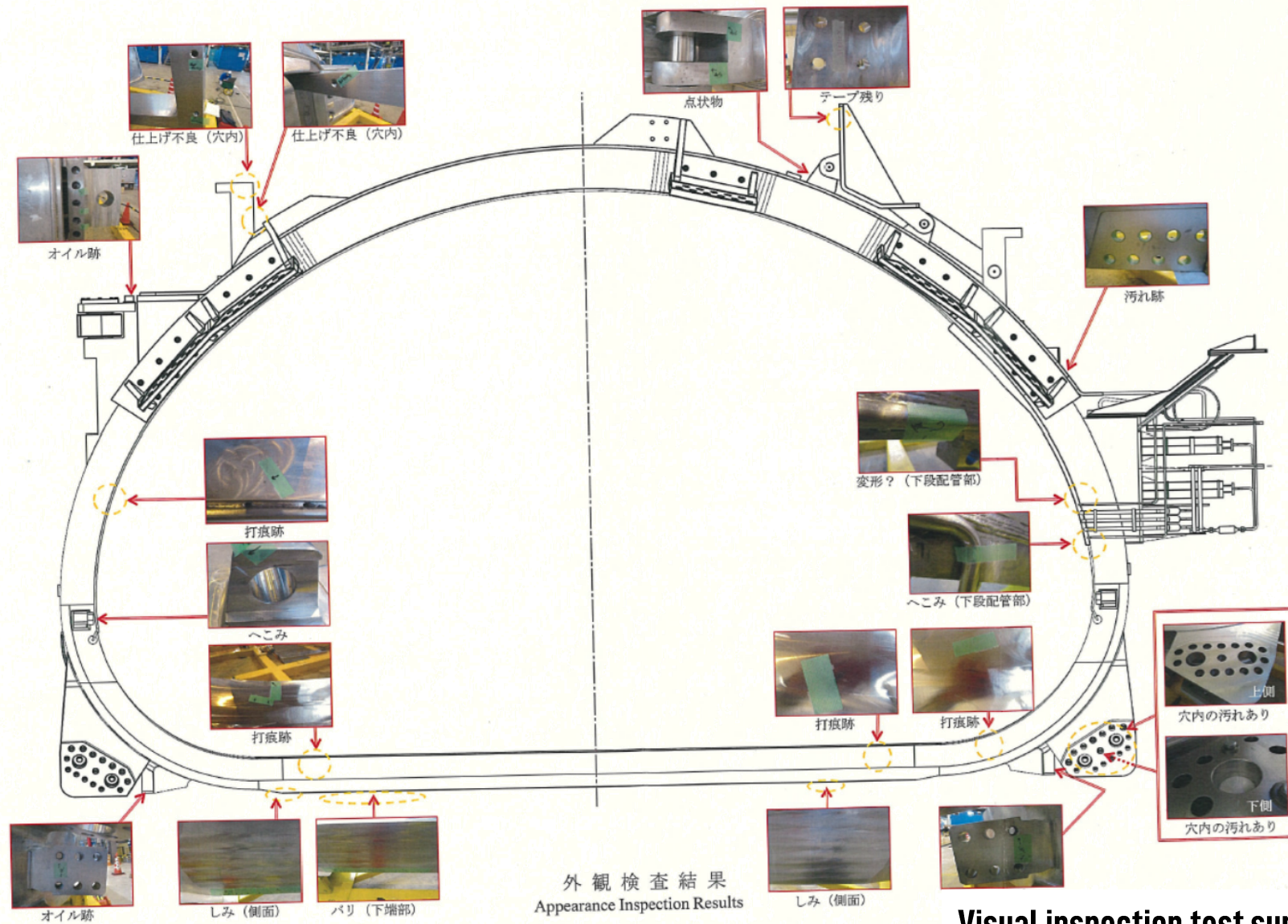
Shock and tilt indicators – if triggered it means that the package and the components have been subject to impact or mishandling



<https://www.shockwatch.com.au/>

Shock and temperature loggers – to keep track of all abnormal events

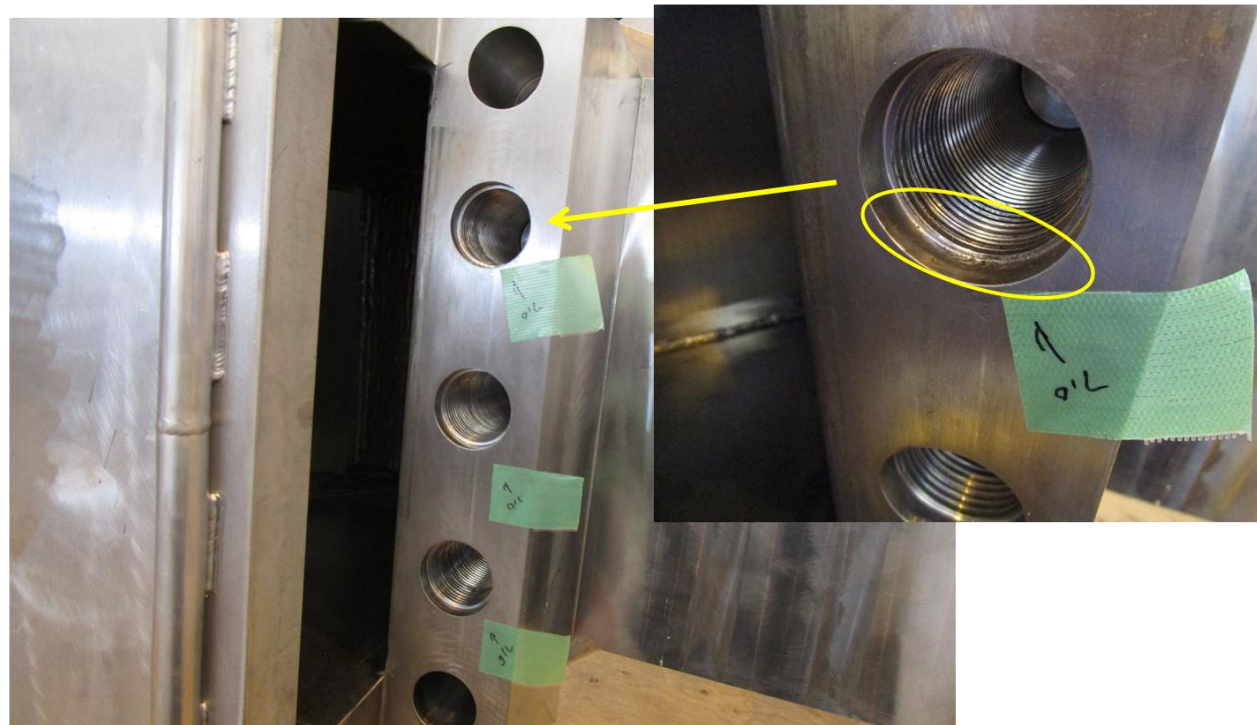
- If the shock indicators and loggers show no issue, the packaging is opened and the components inspected.
- The visual inspection of the component is done to ascertain that there are no deviations from the component cleanliness and finishing requirements, e.g.:
 - Traces of oil or grease
 - Traces of cutting fluid
 - Traces of paints, or markings
 - Dents or scratches on interface surfaces, etc.
 - Foreign material (metal filings, swarf, dust, rust)
- Typically, a comprehensive checklist is prepared, so that the operators can follow it to inspect all critical points



Visual inspection test summary of a JT-60SA TF Coil

Thread holes for GS

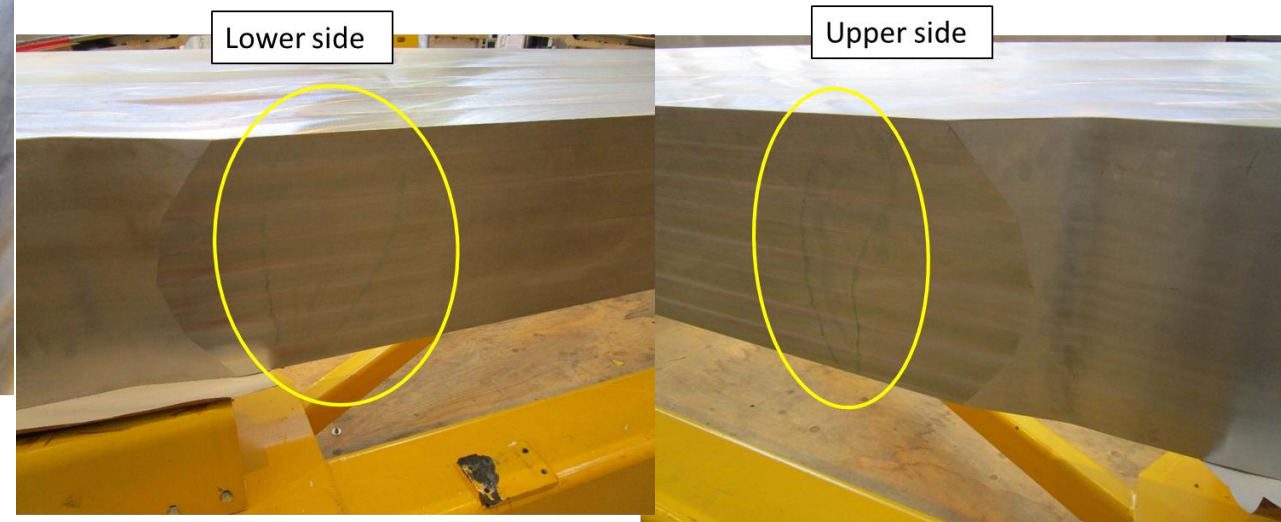
Oil are also remaining on the these thread holes (especially top one).



Oil traces found on a machined hole (cutting fluid residue)

Paints (?) on the inner leg

Could you confirm where these came from ?



Paint traces found on a machined surface

Remaining

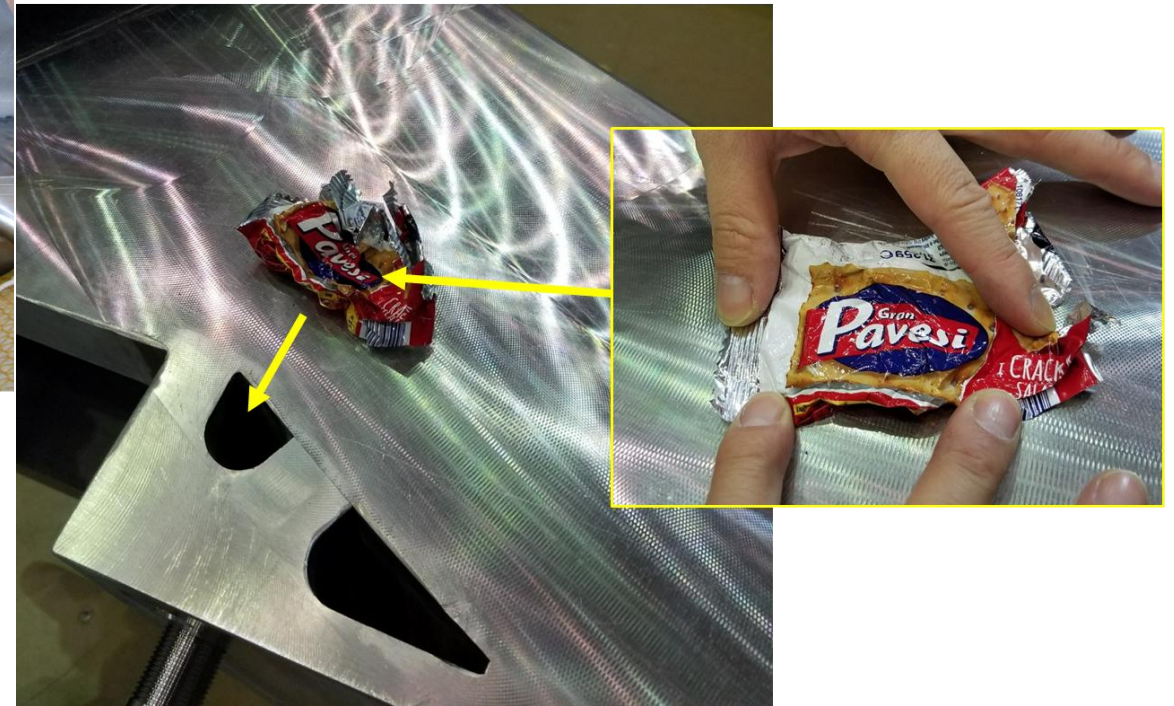
Inside the bag, a roll of tape and a piece of cloth (?) remained around EF1 pedestal.



Some foreign materials might remain inside the packaging

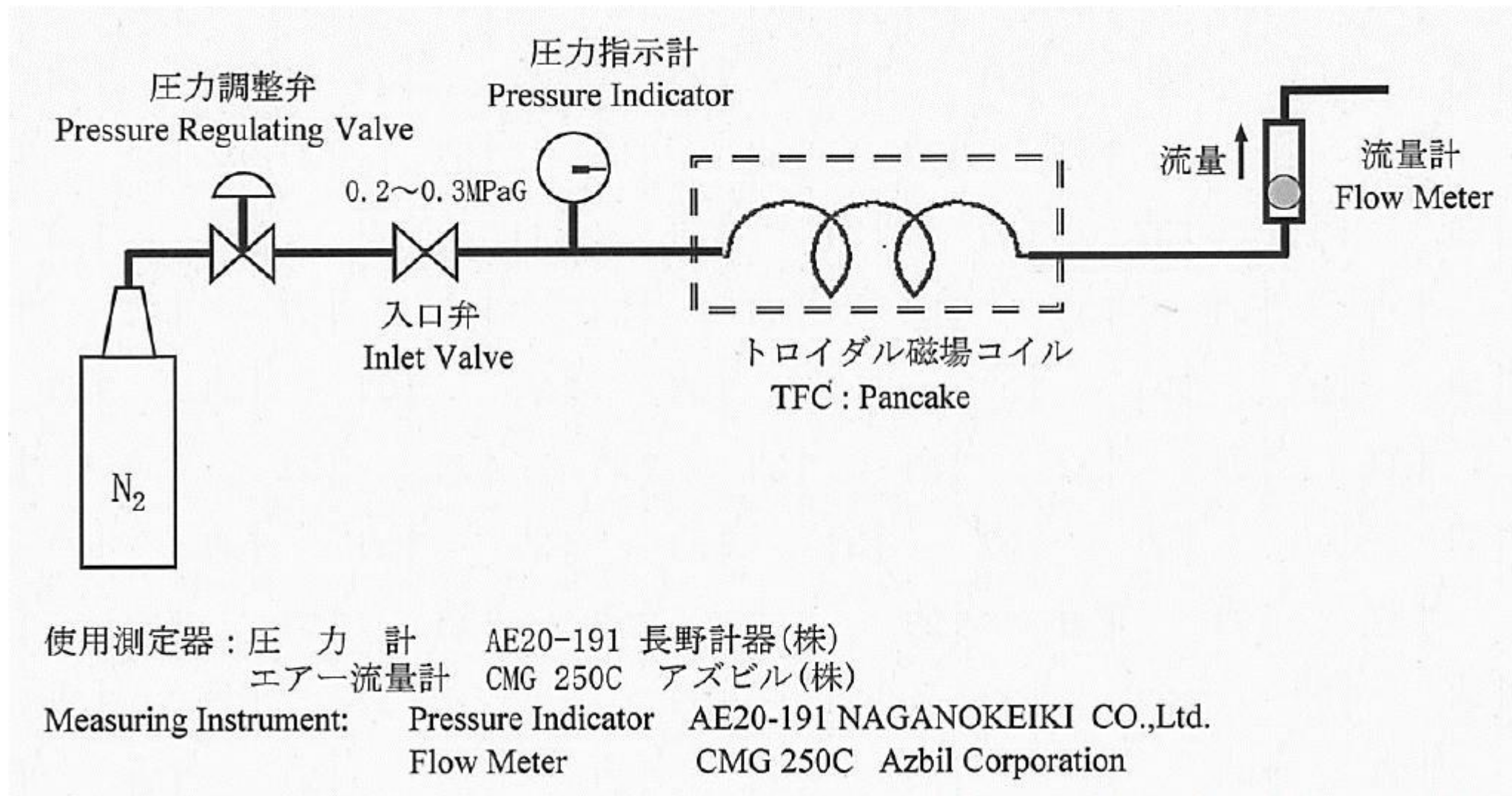
Remaining (2)

An “unbelievable” material was found in the hole of EF5 pedestal !
Please check the reason why this material was involved.



Hidden treasures

- After visual test, more comprehensive testing is performed, to validate some of the key parameters of the components.
- For those components which are actively cooled, pressure, flow and leak tests are usually performed on arrival as incoming inspection tests.
- This is done to ascertain that the packing and shipment procedures have not damaged the component, prior to its connection to the cooling system of the experiment.



A typical pressure and flow test setup



No. 25

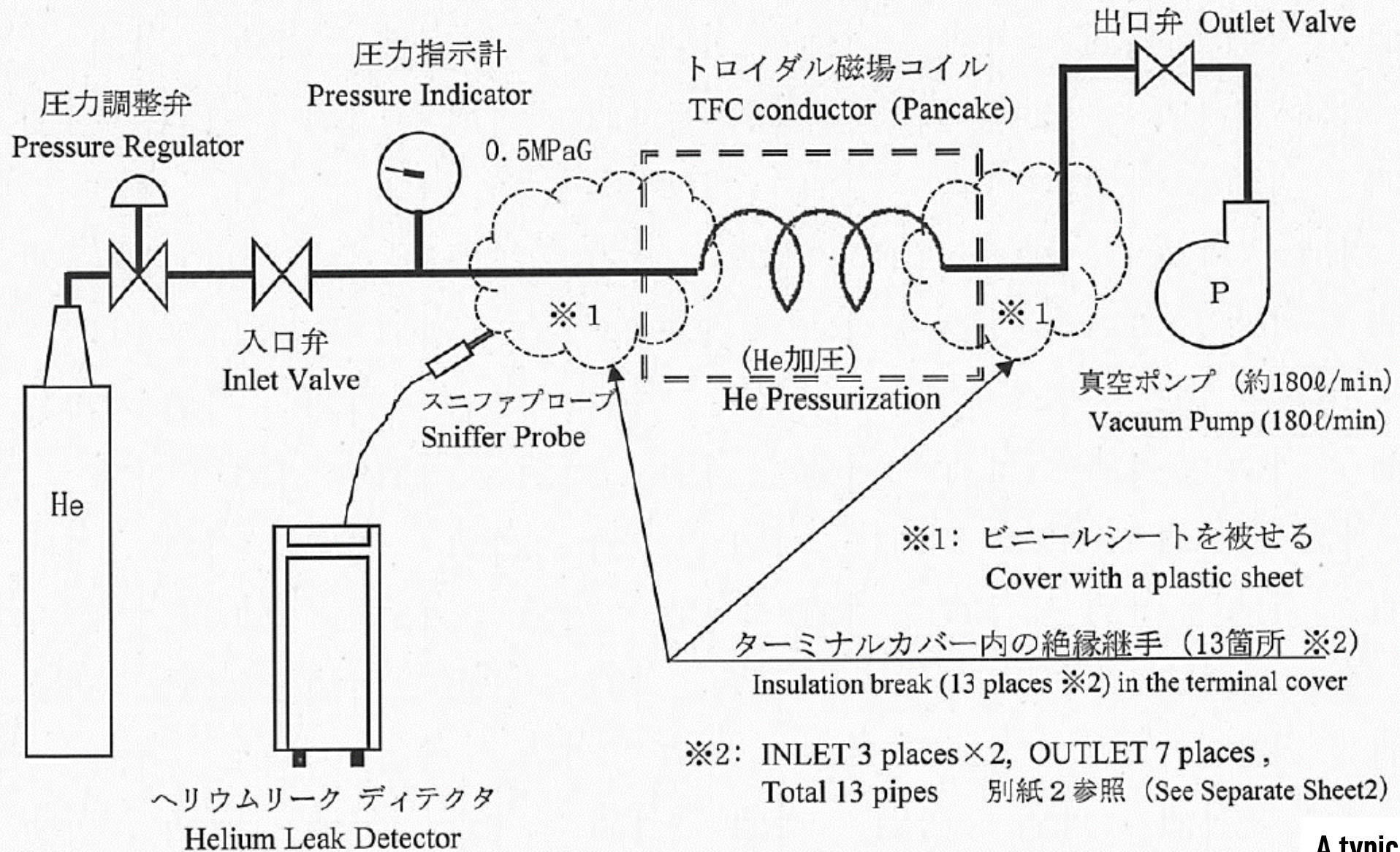
JT-60SA トロイダル磁場コイル
受入検査

2018年2月1日

流路抵抗試験

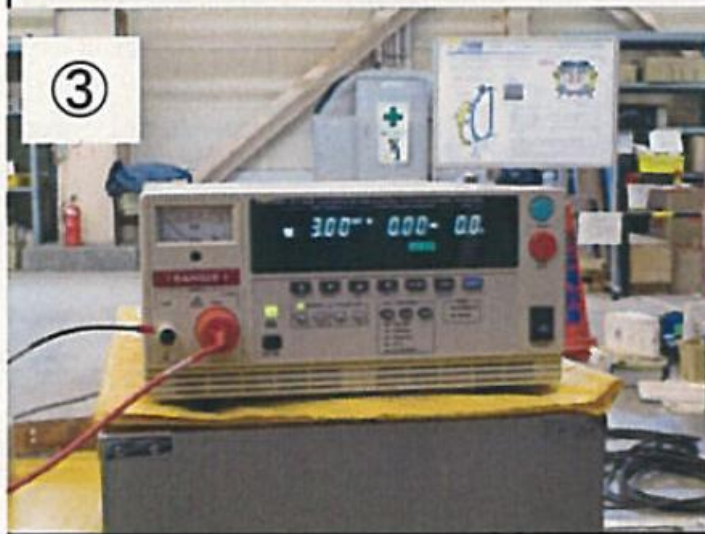
- ① ヘリウムガスの排出(真空ポンプ使用)
 - ② 窒素ガス(N₂)充填(0.20MPaG)
 - ③ 流量計による流量値測定
- 測定結果 : 83ℓ/min

An extract from a flow test report



A typical leak test setup

- For electrical components, like the magnet coils, current leads, joint boxes, etc., electrical tests must be repeated also during incoming inspection.
- This is particularly important, as an electrical fault during operation can jeopardize machine operation and even the overall investment.
- Typically, the following tests are performed:
 - High voltage DC test
 - High voltage AC test
 - Sensor tests
 - Sensor cable continuity and resistance to ground tests



No. 16

JT-60SA トロイダル磁場コイル
受入検査

2018年2月1日

耐電圧試験及び絶縁抵抗試験 (3)

【耐圧試験状況】

① 耐電圧試験器の設定

・印加電圧: DC3kV

・印加時間: 60sec

・判定リーク電流値: 0.02mA (200 μ A)

② 試験状況

③ 試験結果: <10 μ A (0.00mA)

④ 試験終了後の放電作業

An extract from a high voltage test report



No. 17

JT-60SA トロイダル磁場コイル
受入検査

2018年2月1日

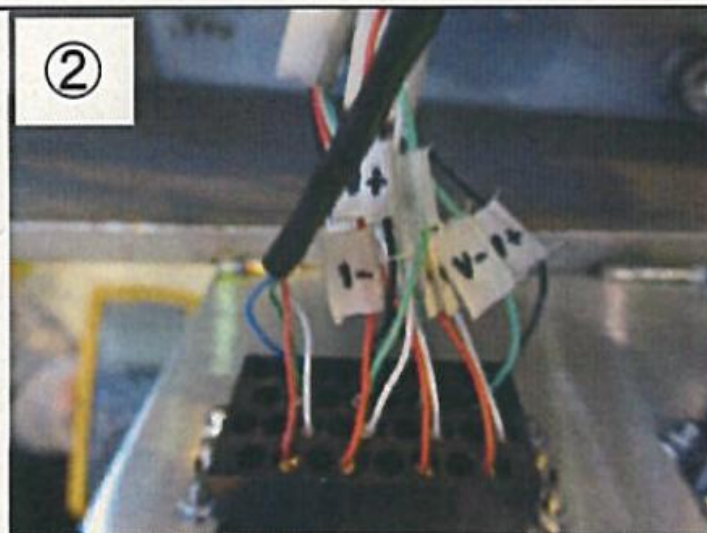
耐電圧試験及び絶縁抵抗試験 (4)

・耐電圧試験後の絶縁抵抗試験

試験結果: 133 GΩ

(印加電圧DC1kV)

An extract from a ground resistance test report



No. 18

JT-60SA トロイダル磁場コイル
受入検査

2018年2月1日

温度センサ導通(健全性確認)検査(1)

① 定電流発生器 No.4($10\mu\text{A}$ 発生)の
健全性確認(金属皮膜抵抗 75Ω 使用)

※確認結果： 75.91Ω

② 温度センサ用コネクタと配線状況

③、④ 確認検査状況

An extract from a temperature sensor test report



No. 22

JT-60SA トロイダル磁場コイル
受入検査

2018年2月1日

電圧タップ用4芯ケーブルの導通及び
絶縁抵抗検査(2)

・各線とコイルケース間の絶縁確認検査

①～② 検査結果：全てOL(50Ω以上)

An extract from a sensor cable ground resistance test report

- For all components which need to be assembled precisely in the experiment, key geometrical dimensions must be verified on arrival.
- This is necessary to guarantee that once the component is moved to the assembly hall, there will be no 'surprises'. So the component is measured using laser metrology while still on its transportation frame.
- Only the most important dimensions are verified at this stage. To be noted that all components leave the factory with a complete QA documentation including also the geometrical dimensions. The metrology at the incoming inspection is done to exclude deformations occurred during transport.



No. 11

JT-60SA トロイダル磁場コイル
受入検査

2018年3月5日

レーザートラッカによる寸法検査(2)
・リフレクタによるレーザートラッカ
調整作業

An extract from a metrology test report



No. 13

JT-60SA トロイダル磁場コイル
受入検査

2018年3月5日

レーザートラッカによる寸法検査(4)

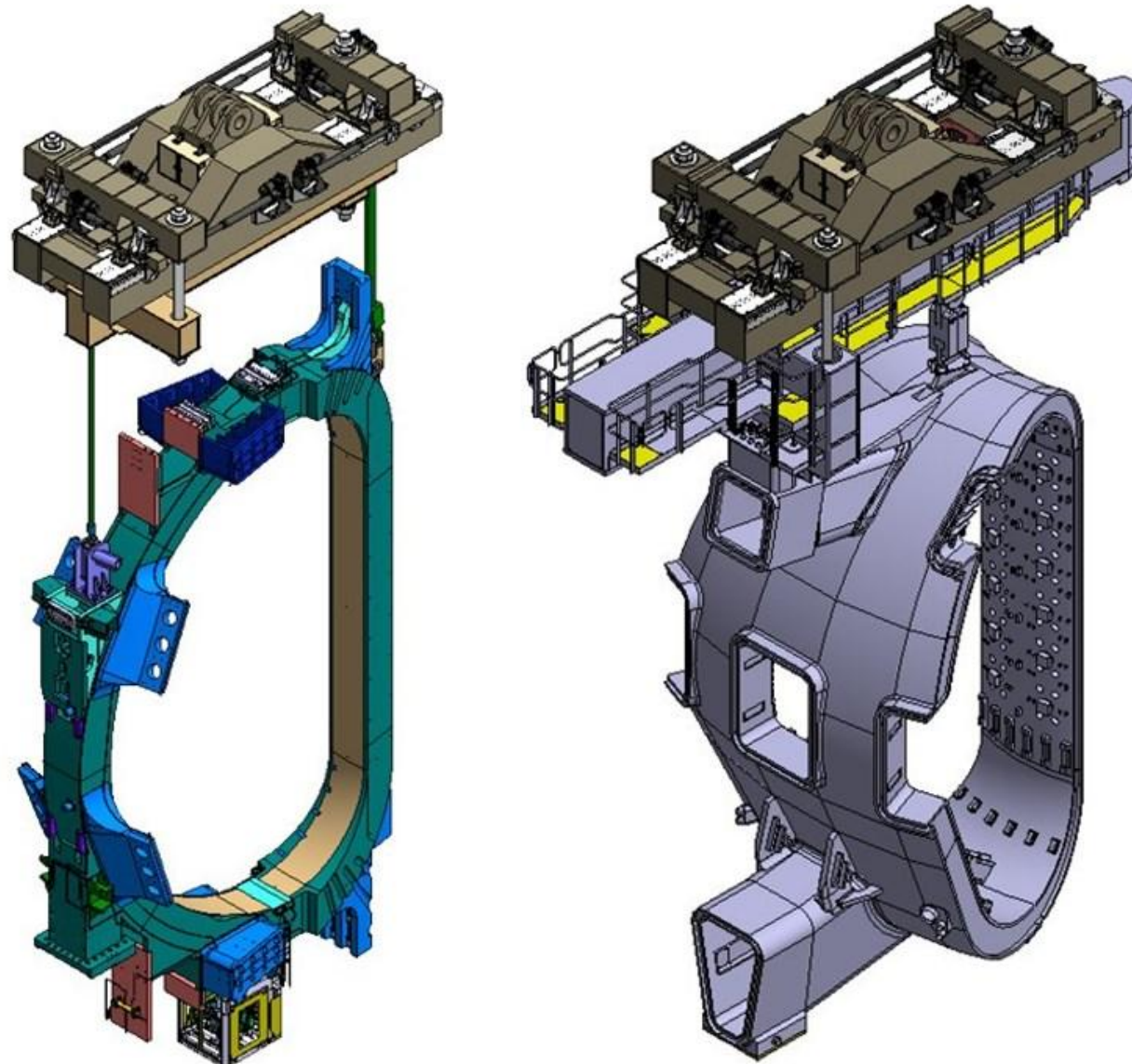
- ① 測定点Qの座標測定
- ② 測定点Fの座標測定
- ③ 測定点Gの座標測定
- ④ 測定点Eの座標測定

An extract from a metrology test report

- Once the components have been inspected, they are ready to be hauled to the assembly hall for their installation in the experiment.
- For large experiments, like ITER, precise and safe handling of components requires maximum effort and customized solutions.
- To be noted that, especially for fusion experiments components, the ratio of weight to volume is particularly high. That is, there are several components which need to be moved in a single lift, and which have very large weight (1000 t and more).
- In more conventional engineering, even if the plants themselves are much larger than e.g. the ITER site, the single components to be assembled are generally lighter.
- For ITER very specific lifting, uplifting and assembly tools have been developed.



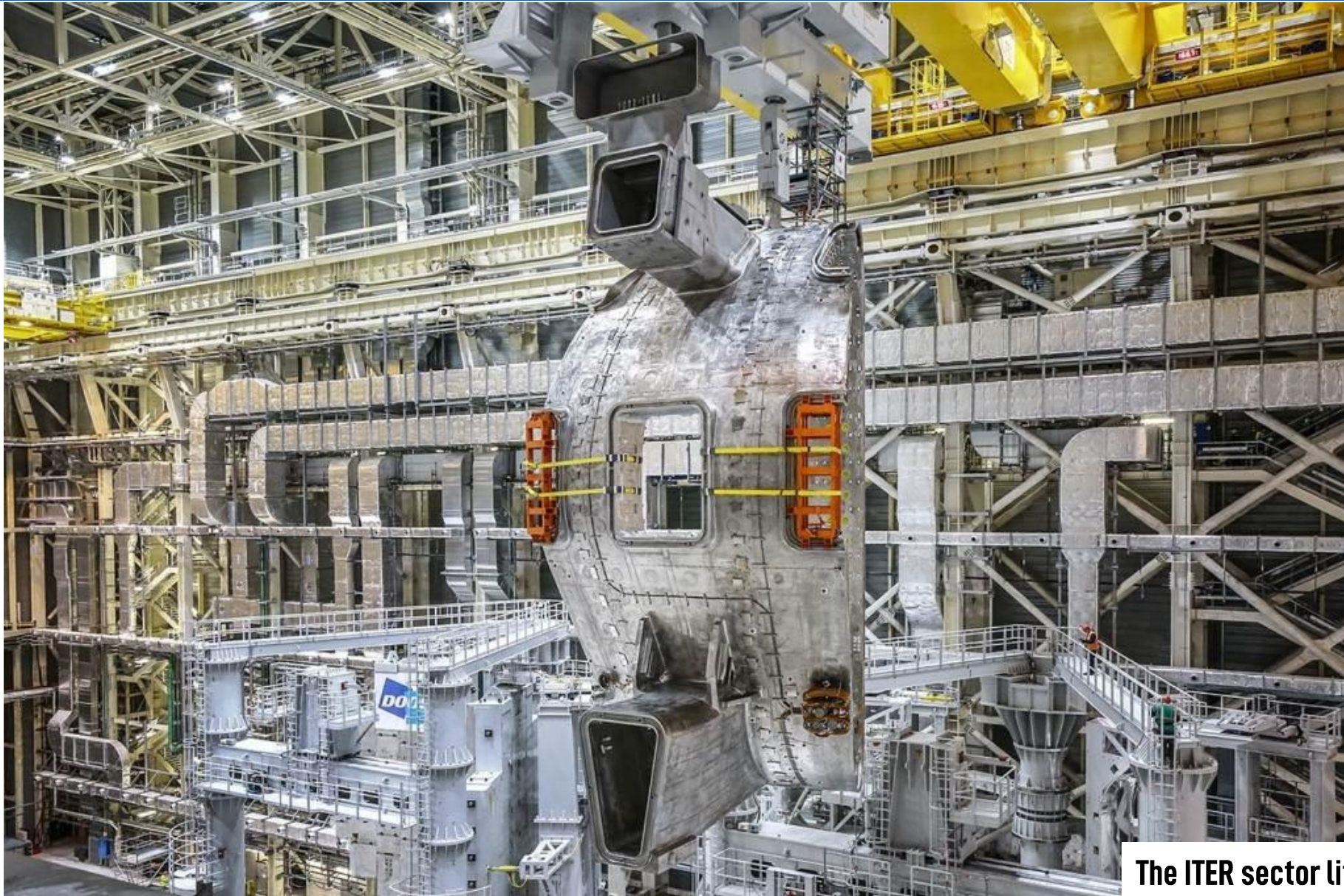
The ITER sector lifting tool



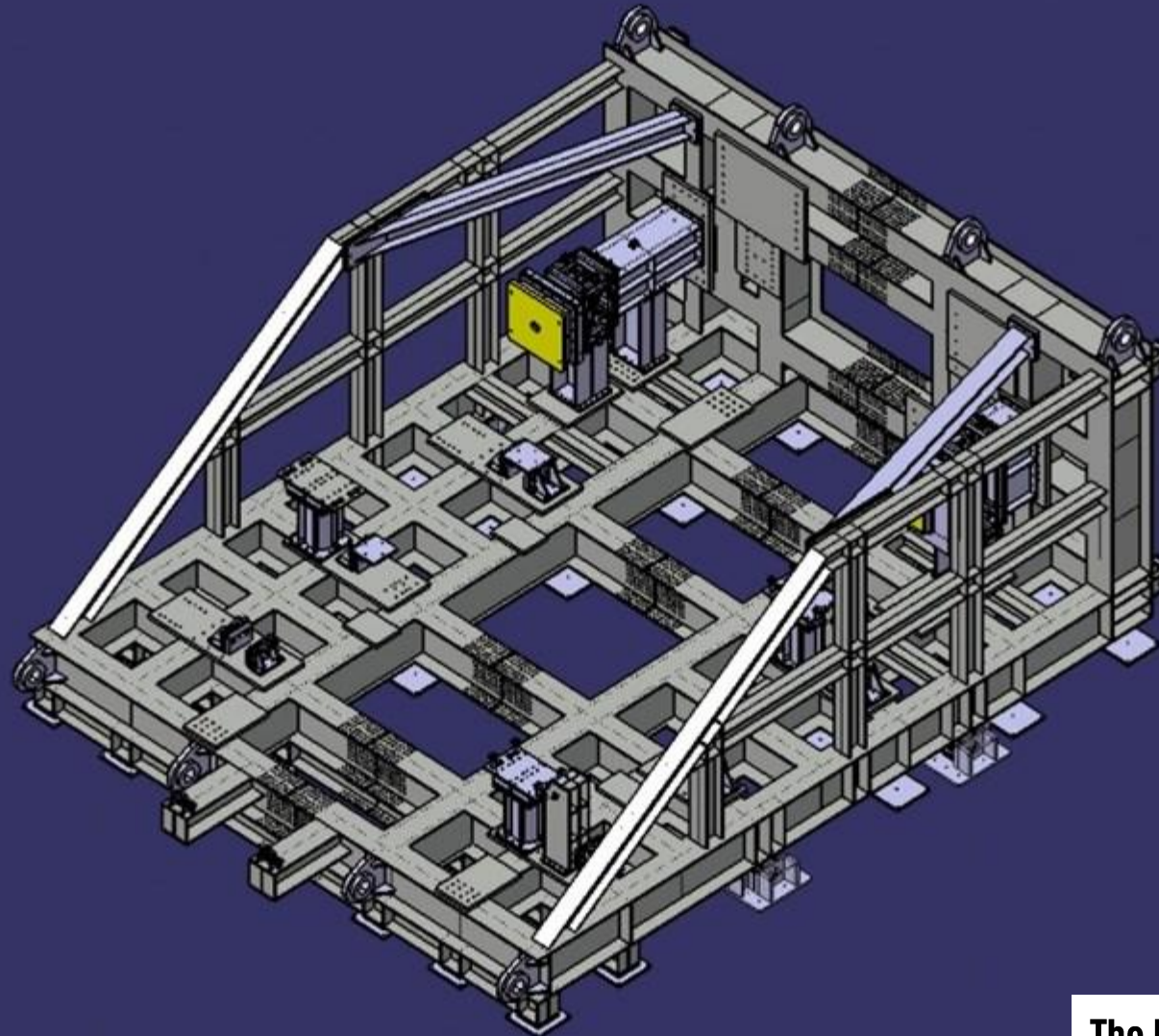
The ITER sector lifting tool



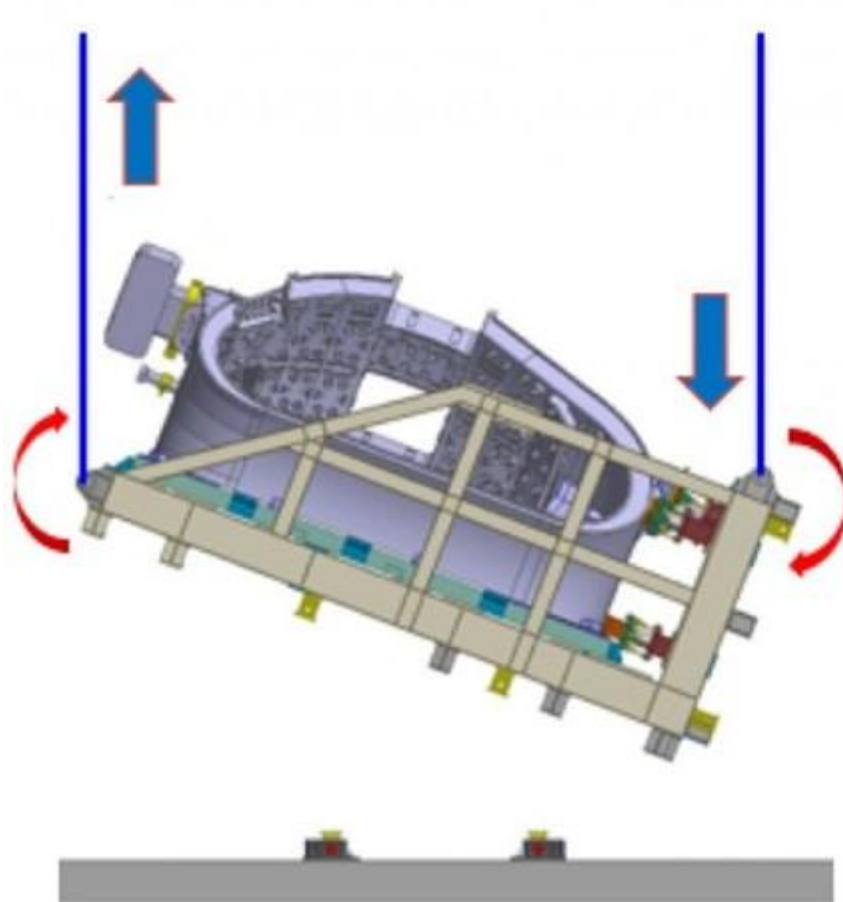
The ITER sector lifting tool



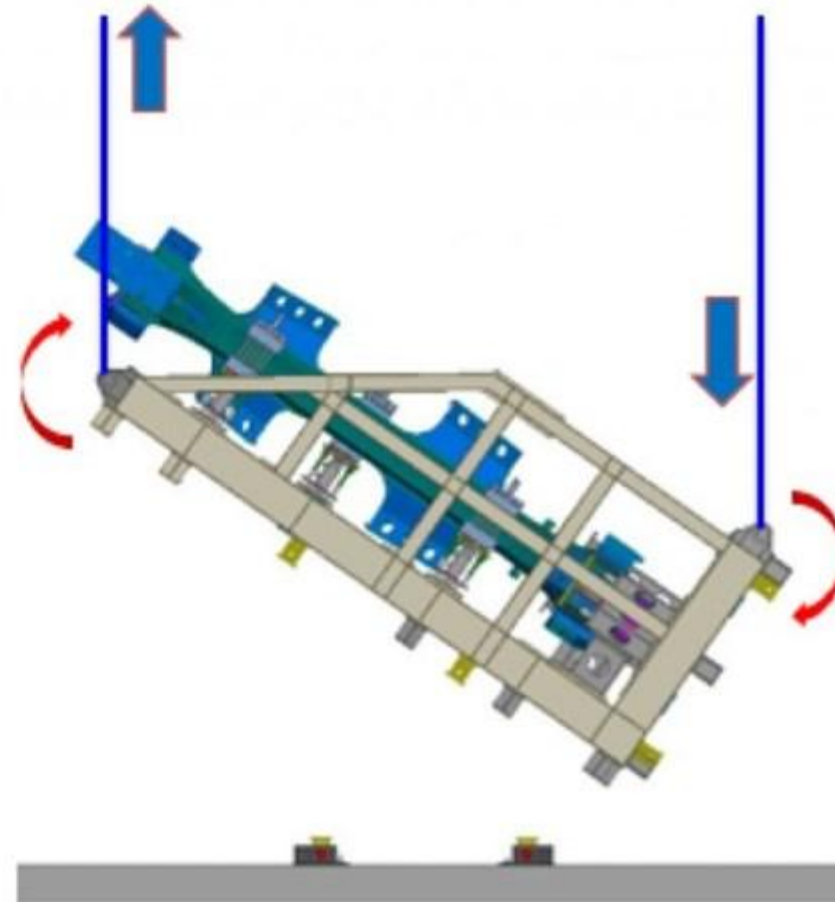
The ITER sector lifting tool



The ITER uplifting tool

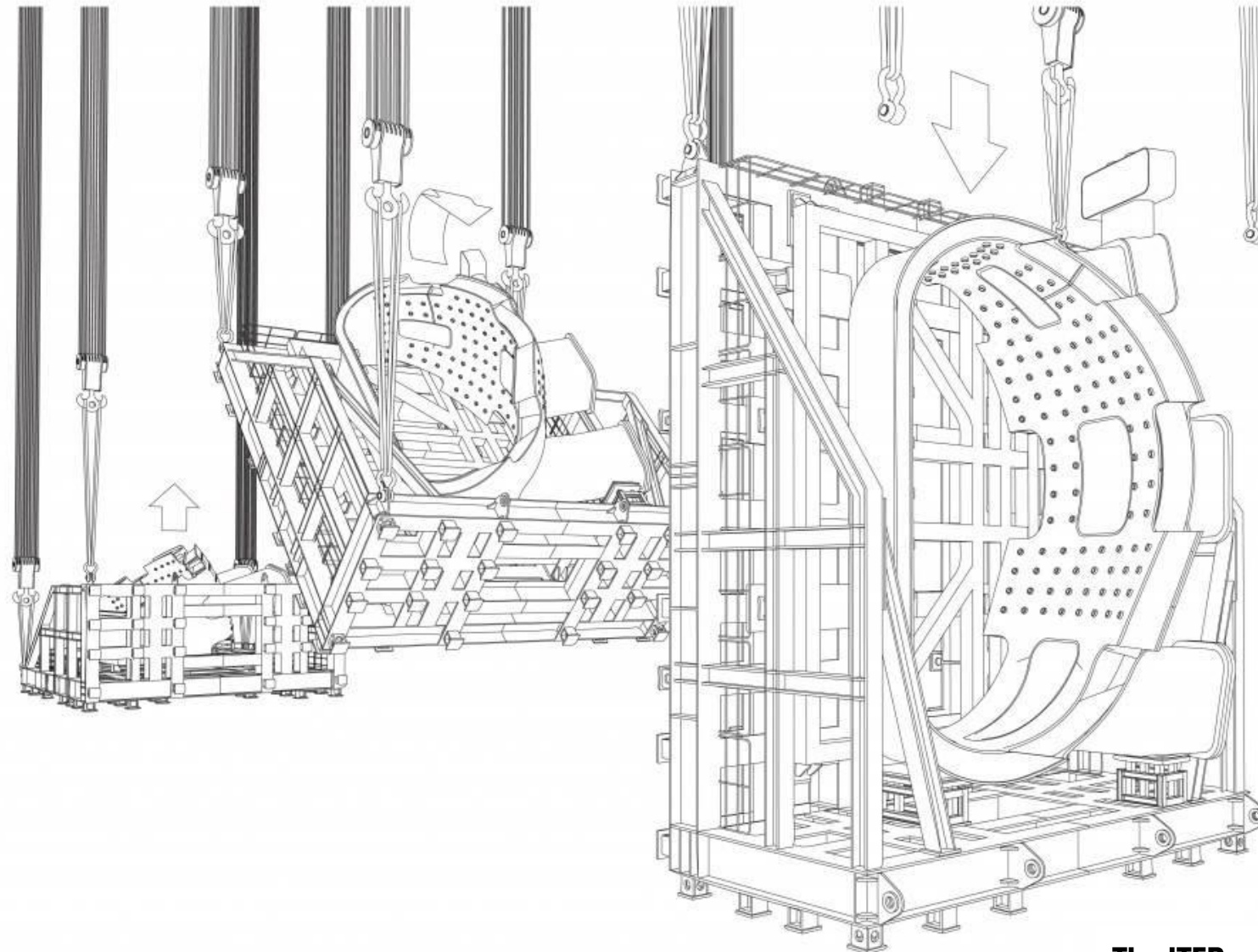


*VV Upending Strategy by 2 x 750
tons Crane Operation*



*TFC Upending Strategy by 2 x 750
tons Crane Operation*

The ITER uplifting tool



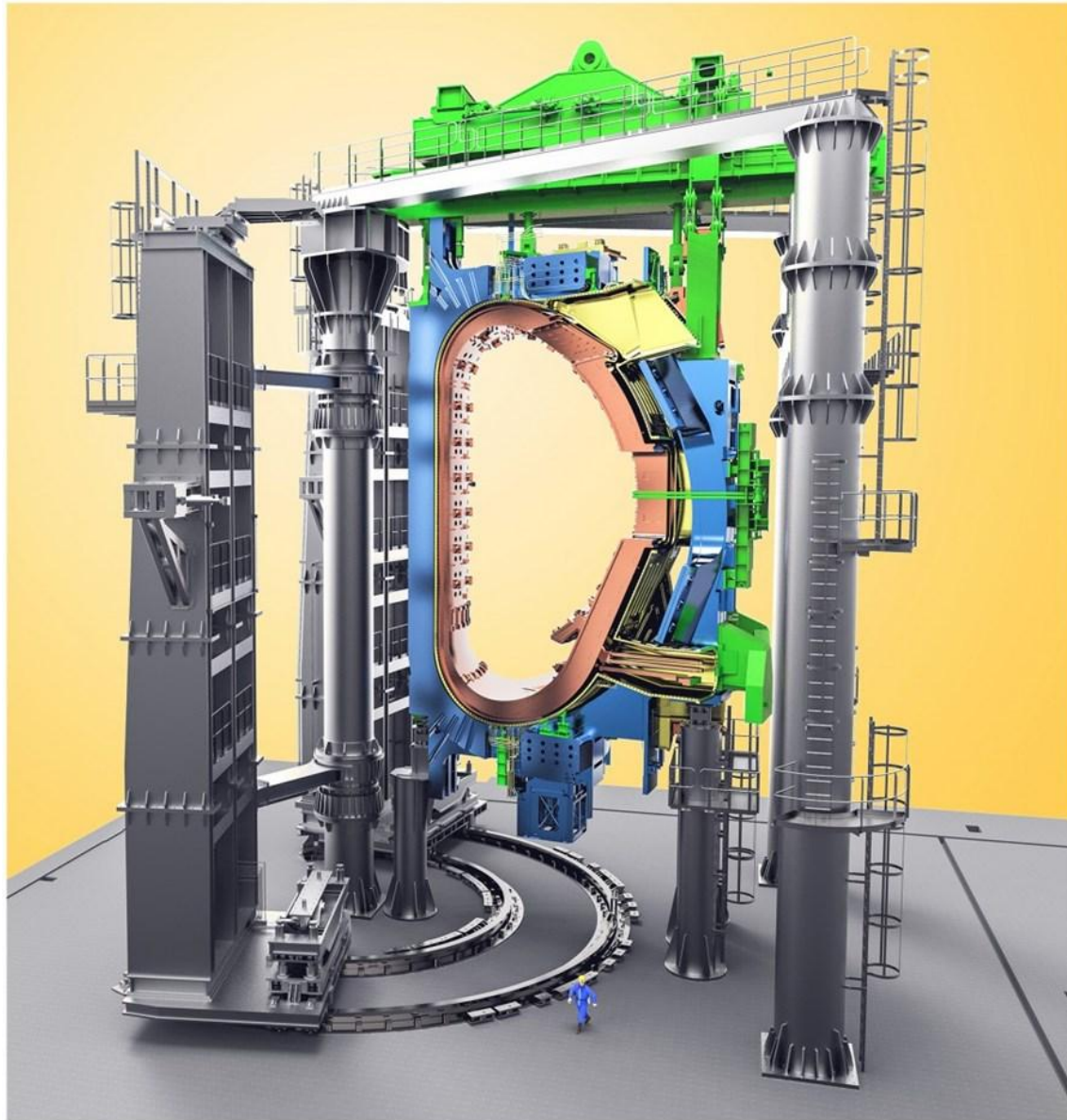
The ITER uplifting tool



The ITER uplifting tool



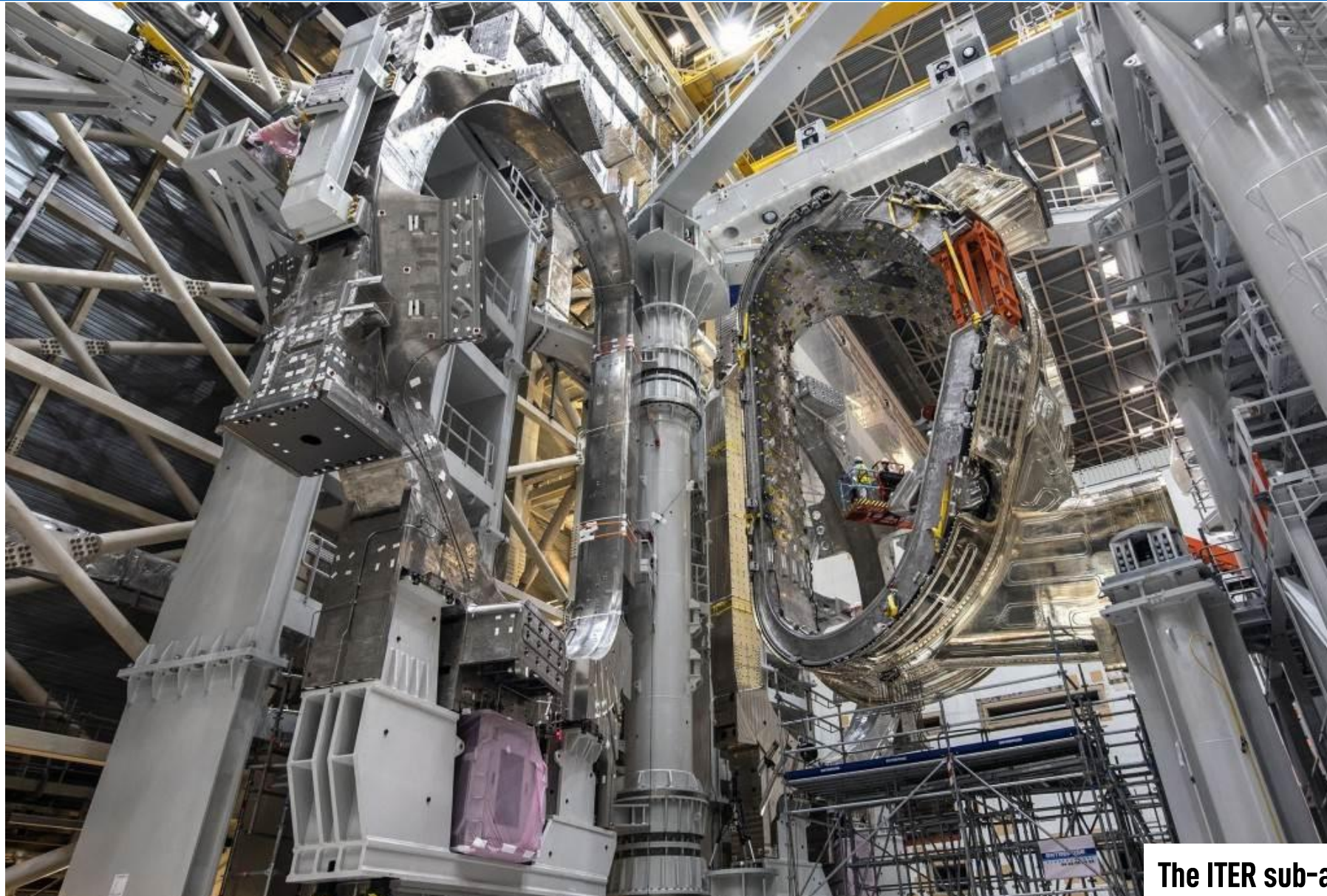
The ITER uplifting tool



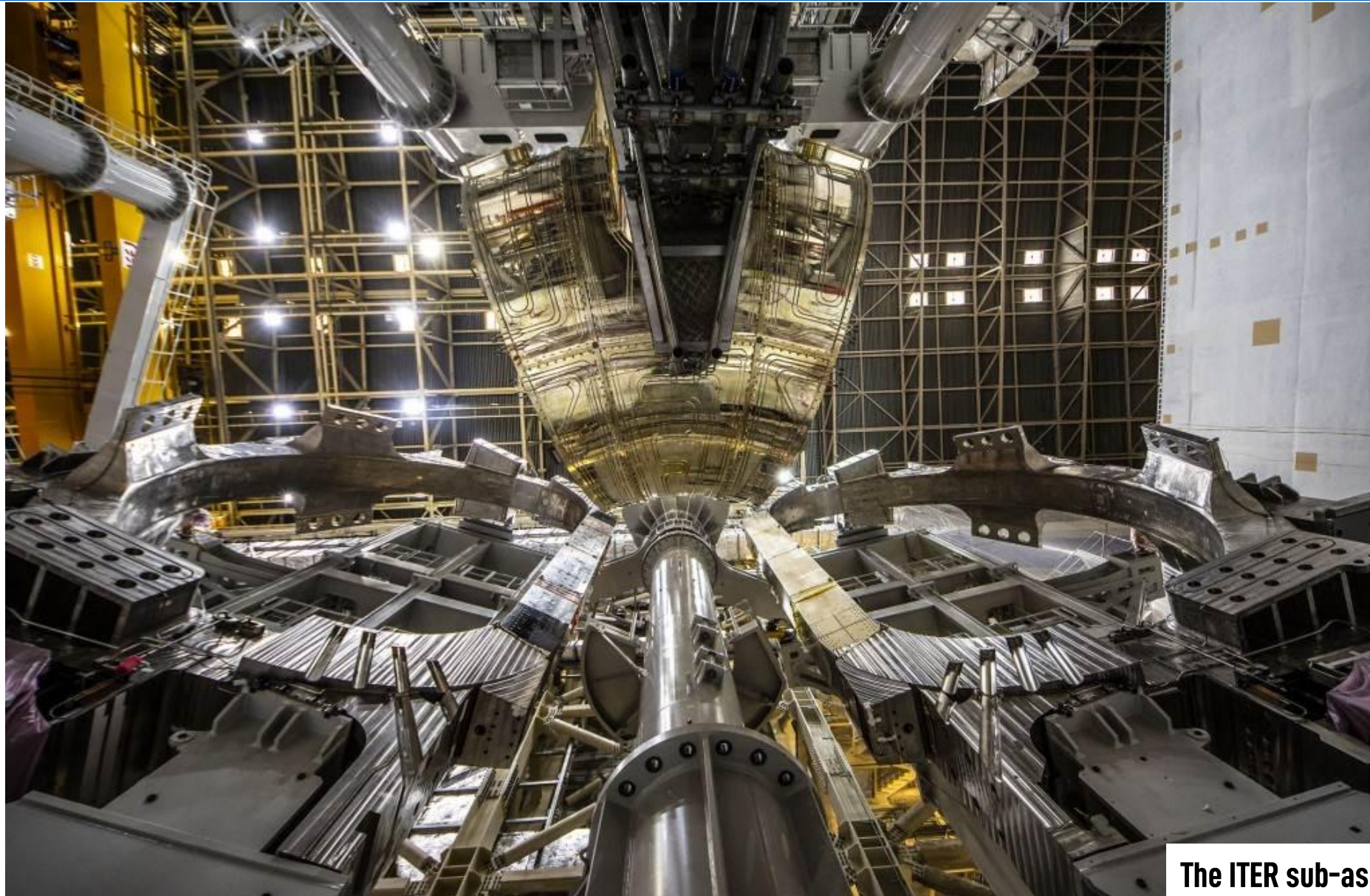
The ITER sub-assembly tools



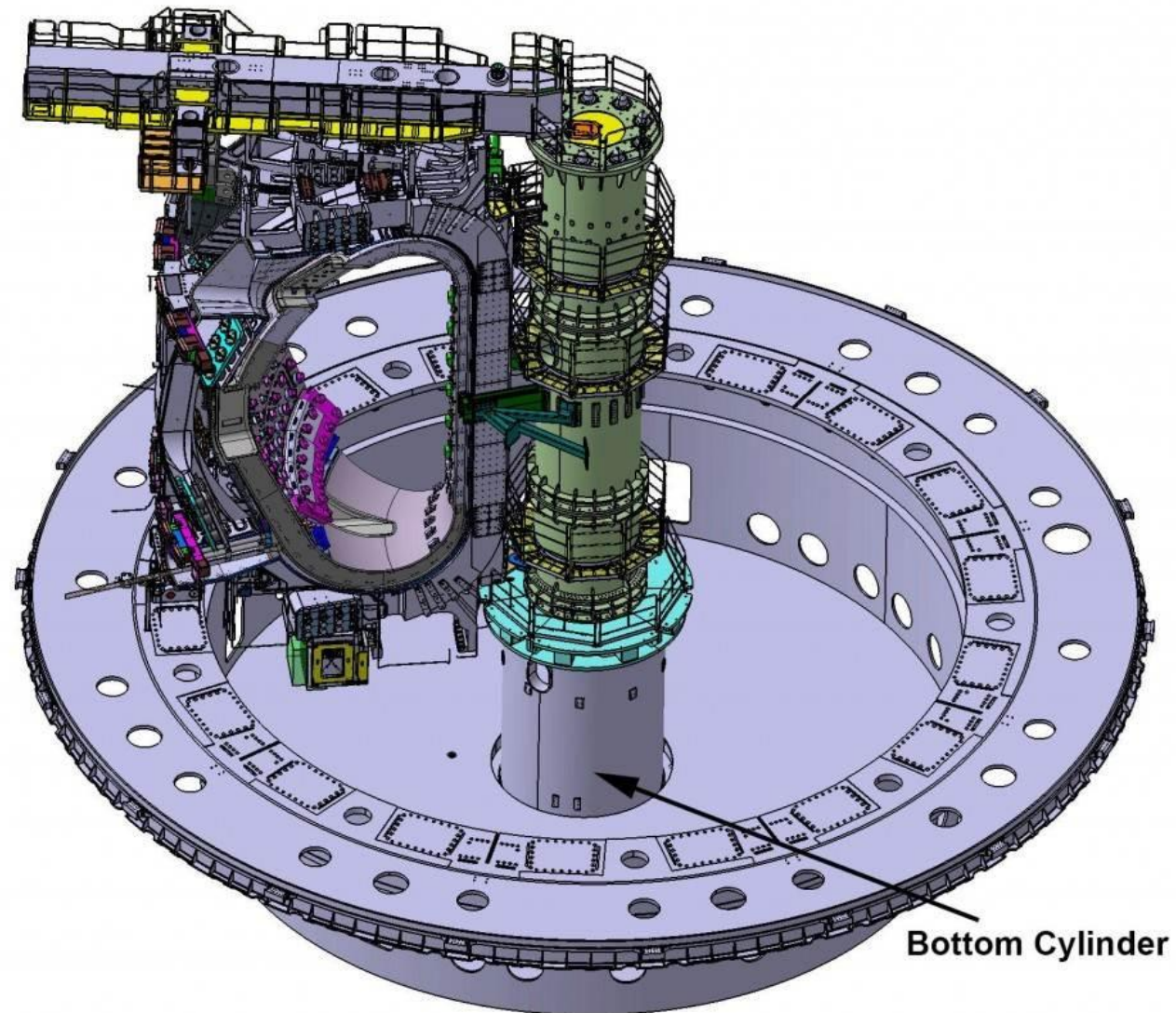
The ITER sub-assembly tools



The ITER sub-assembly tools



The ITER sub-assembly tools



The ITER in-pit assembly tools



The ITER in-pit assembly tools



A complete module (vacuum vessel, thermal shield + 2 TF coils) being moved to the pit



Total weight = 1380 t

A complete module (vacuum vessel, thermal shield + 2 TF coils) being moved to the pit



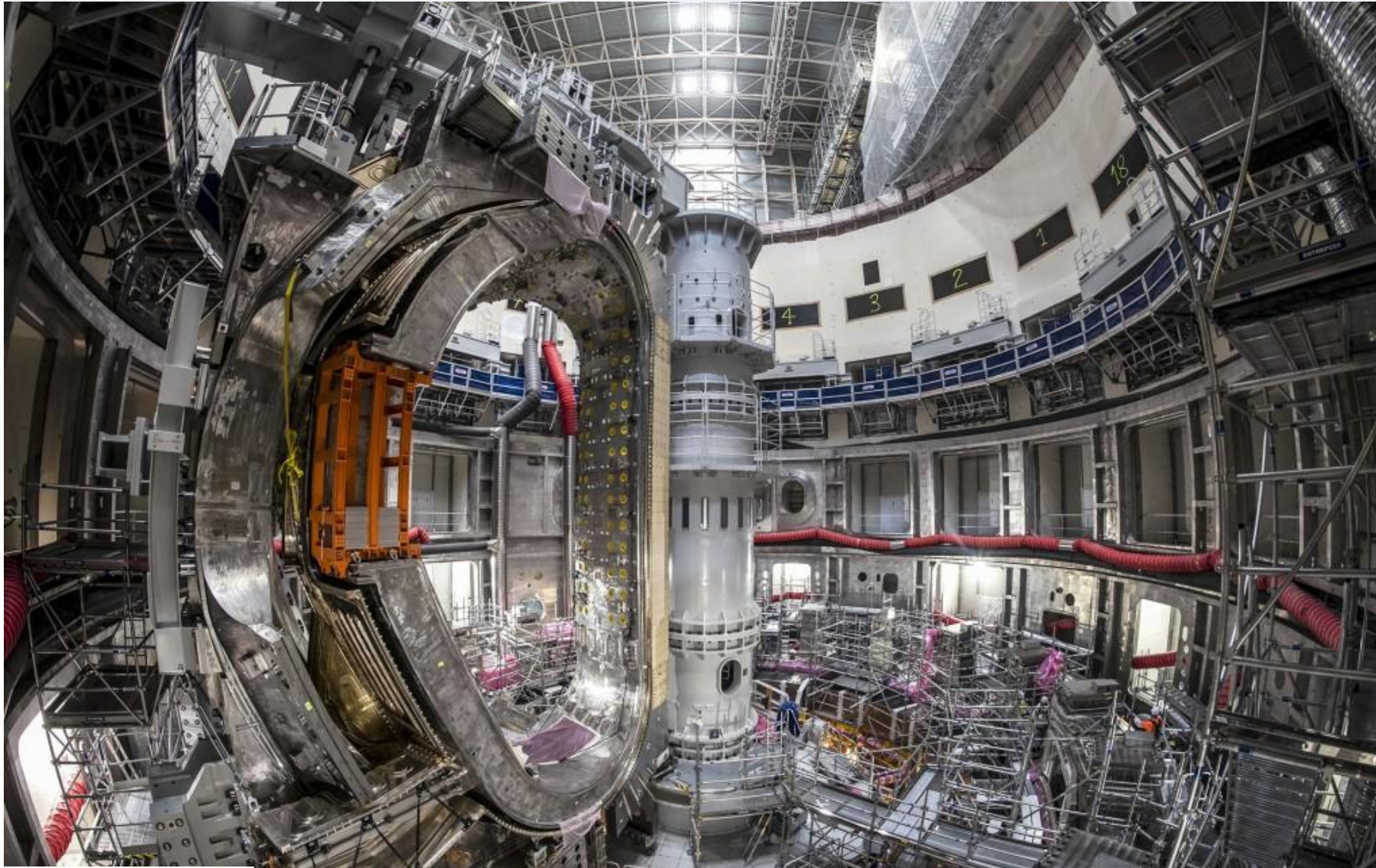
Total weight = 1380 t

A complete module (vacuum vessel, thermal shield + 2 TF coils) being moved to the pit



Total weight = 1380 t

A complete module (vacuum vessel, thermal shield + 2 TF coils) being lowered into the pit



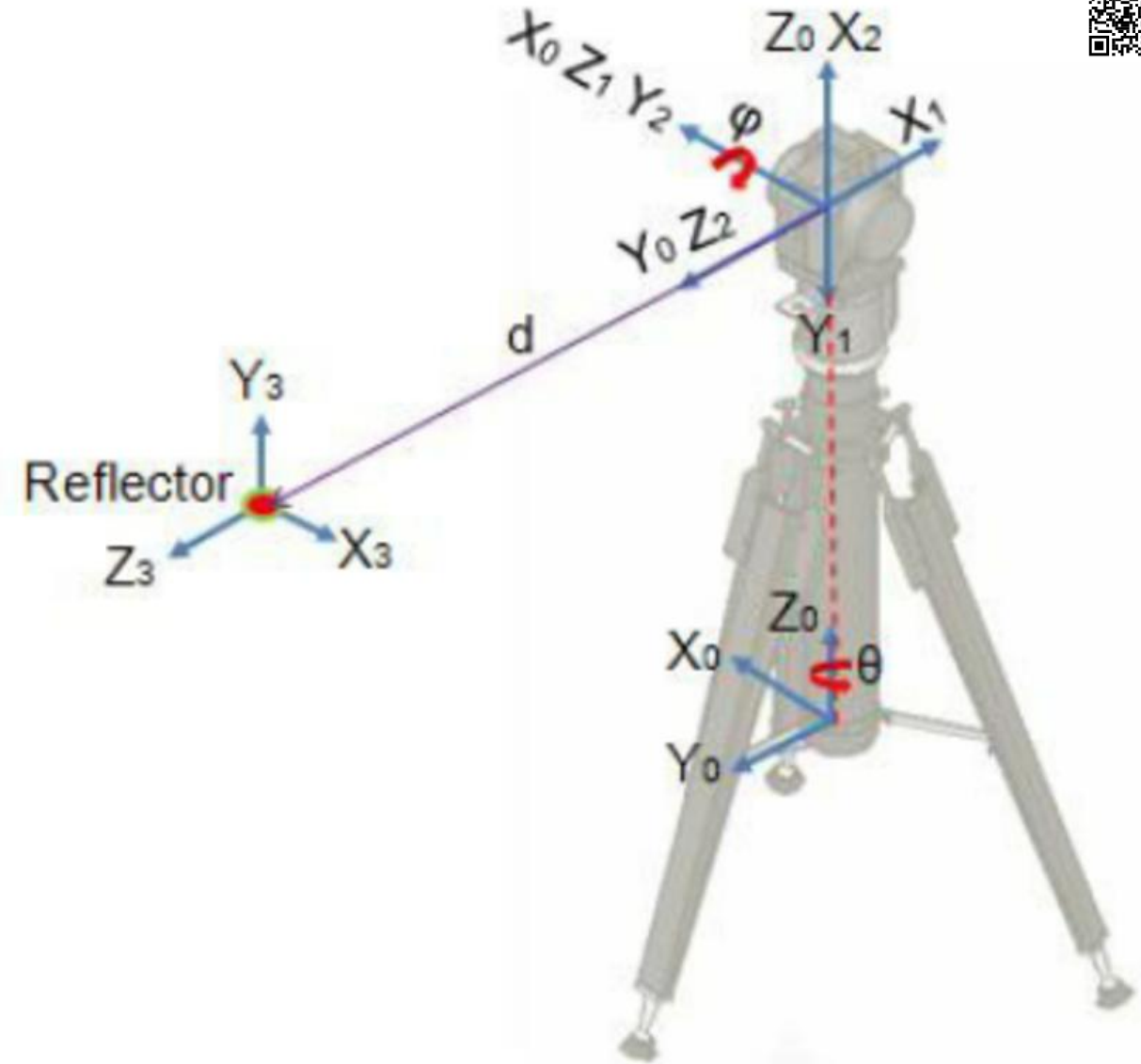
Total weight = 1380 t

A complete module (vacuum vessel, thermal shield + 2 TF coils) close to its final position

- To position such huge components, and to monitor their shape during lifting and in their final position, metrology is used routinely to perform geometrical checks.
- Metrology is the science of measurement, and give us several tools for validating the assembly work in fusion devices. A few examples are:
 - Laser tracking
 - Laser scanning
 - Photogrammetry
- All these methods can be used, together or individually to verify the correct position of the installed components, and to monitor their deformations during installation and operation.



- A laser tracker is a device that can measure distances and angles with very high precision and accuracy (10 microns and more).
- A tracker measures optically the distance between its laser source and a special reflector, and also the 'head' angles of rotation with respect to its base.
- Therefore a tracker can be used to measure points on the surface of an object with very high precision, provided that also stationary reference points are measured together with the object to be measured.





A typical laser tracker



Laser tracker accessories, reflectors

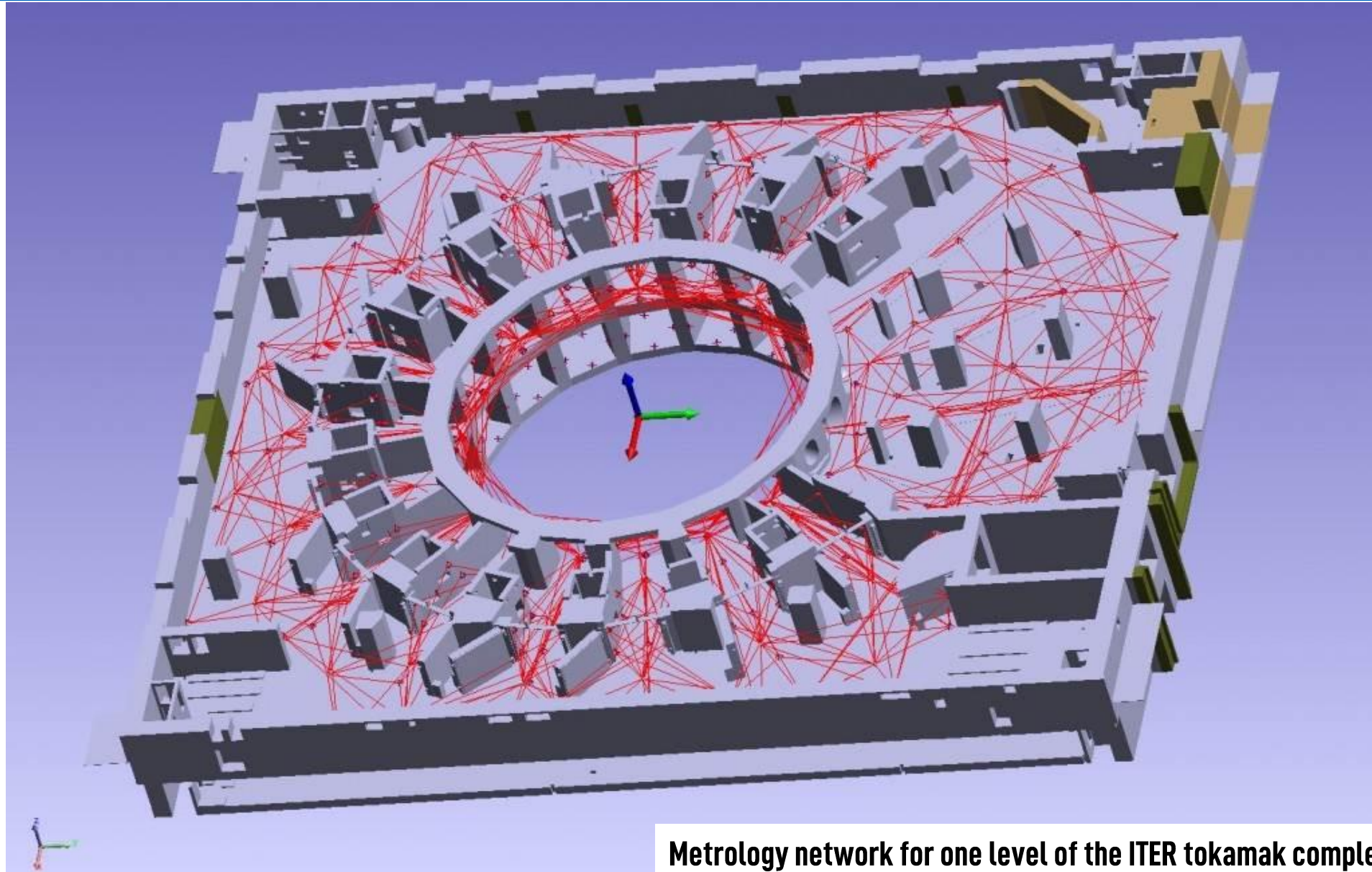
- A laser scanner works in a similar way, but collects point clouds, by repeating the distance measurement many times per second and on large solid angles.
- The obtained point clouds are almost as precise as the laser tracker ones, and require no reflectors.
- Laser scanners can be used to quickly reconstruct complex surfaces in 3D, to use them in digital virtual mockups.
- In this way it is possible to 'test' an assembly process prior to moving the components themselves.



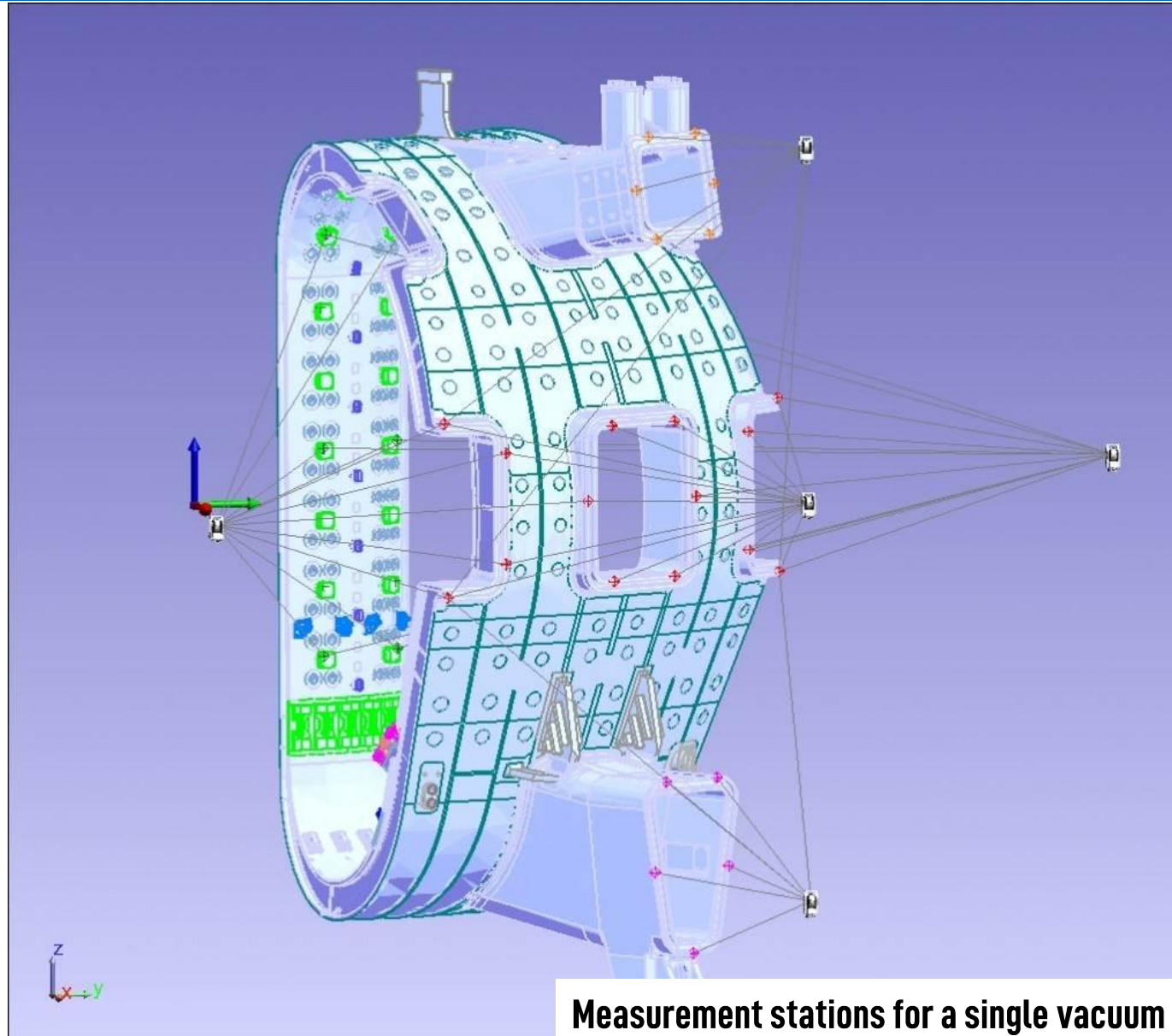
A typical laser scanner



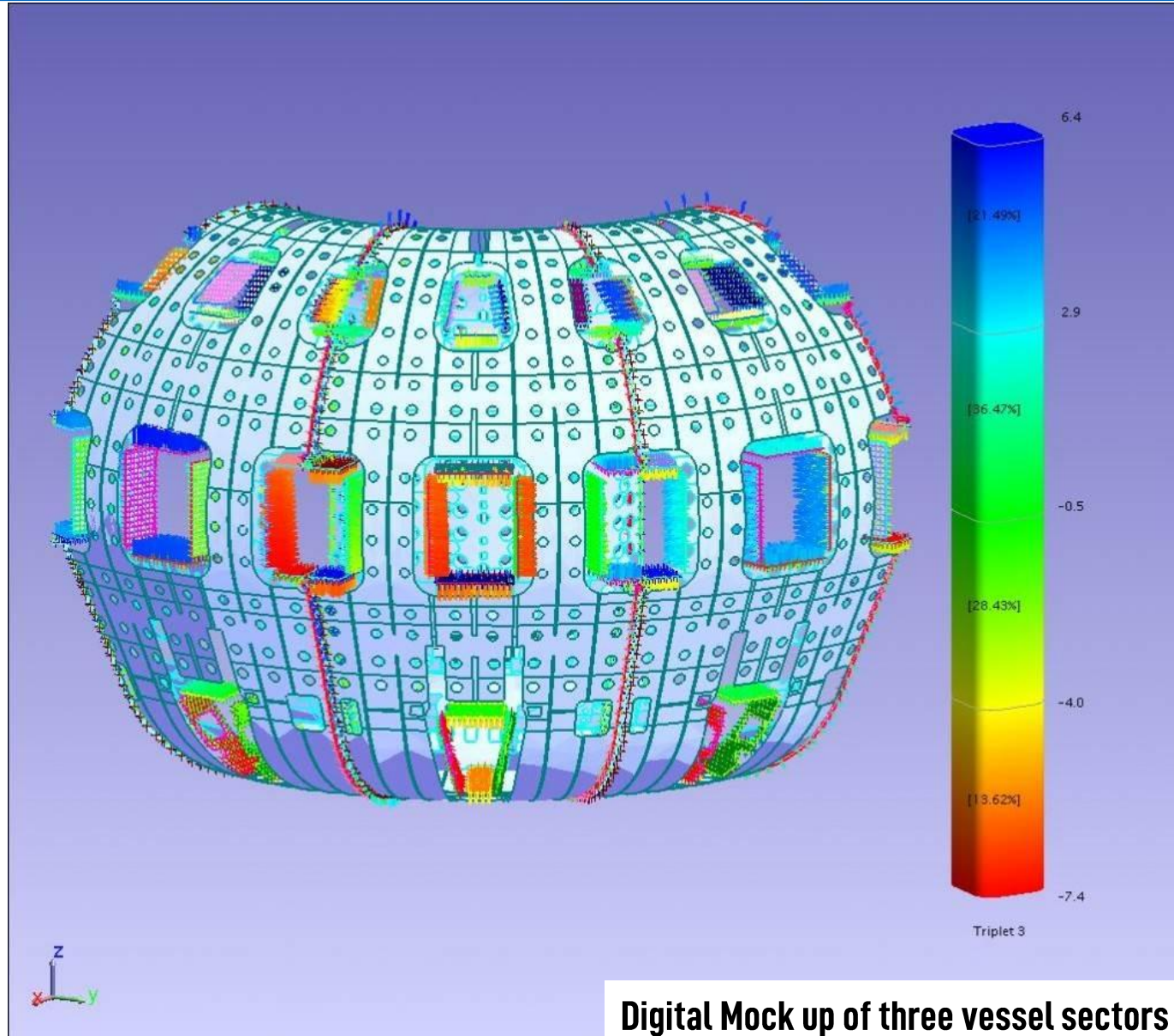
Laser tracker in action



Metrology network for one level of the ITER tokamak complex

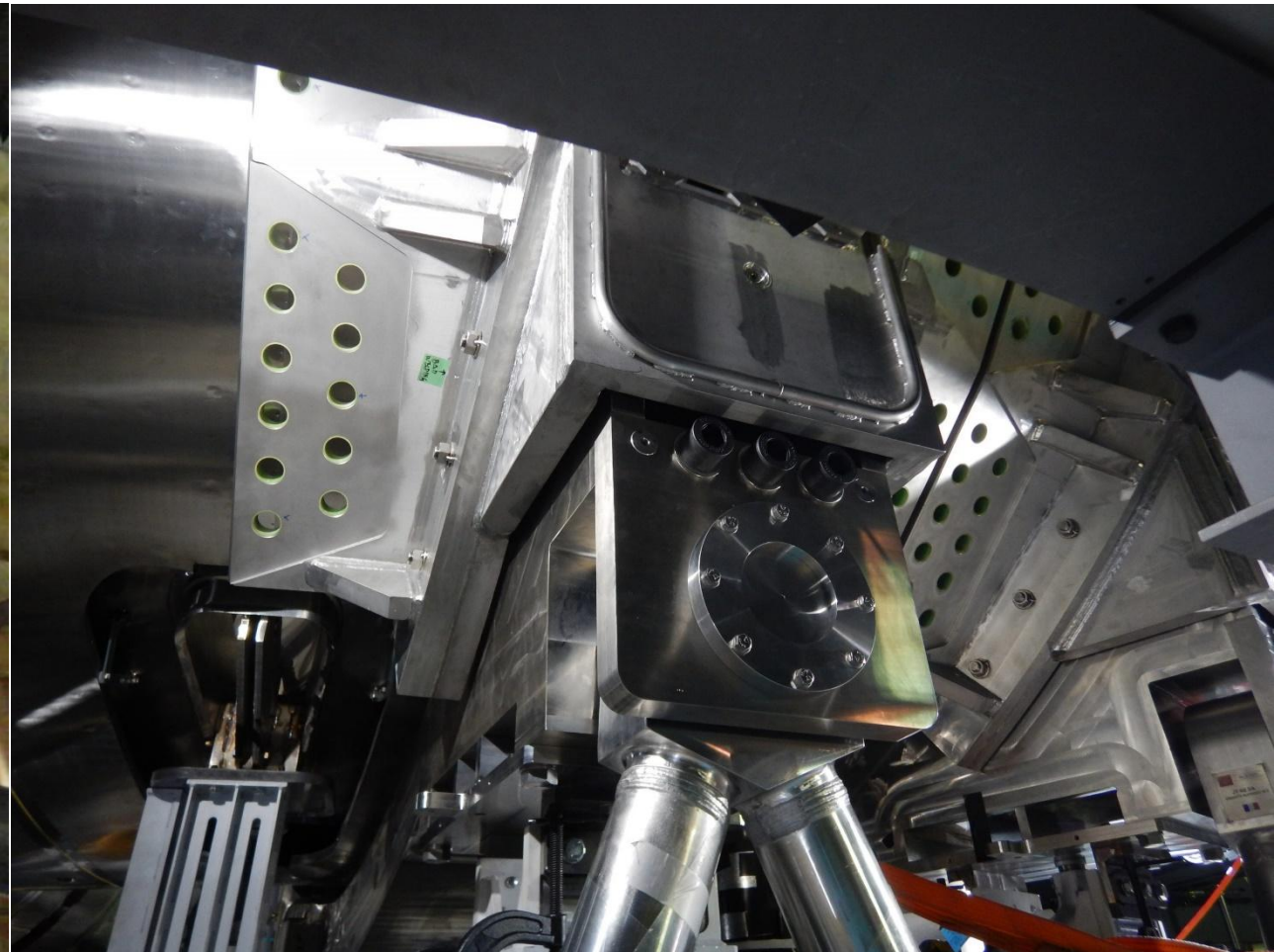


Measurement stations for a single vacuum vessel sector



Digital Mock up of three vessel sectors

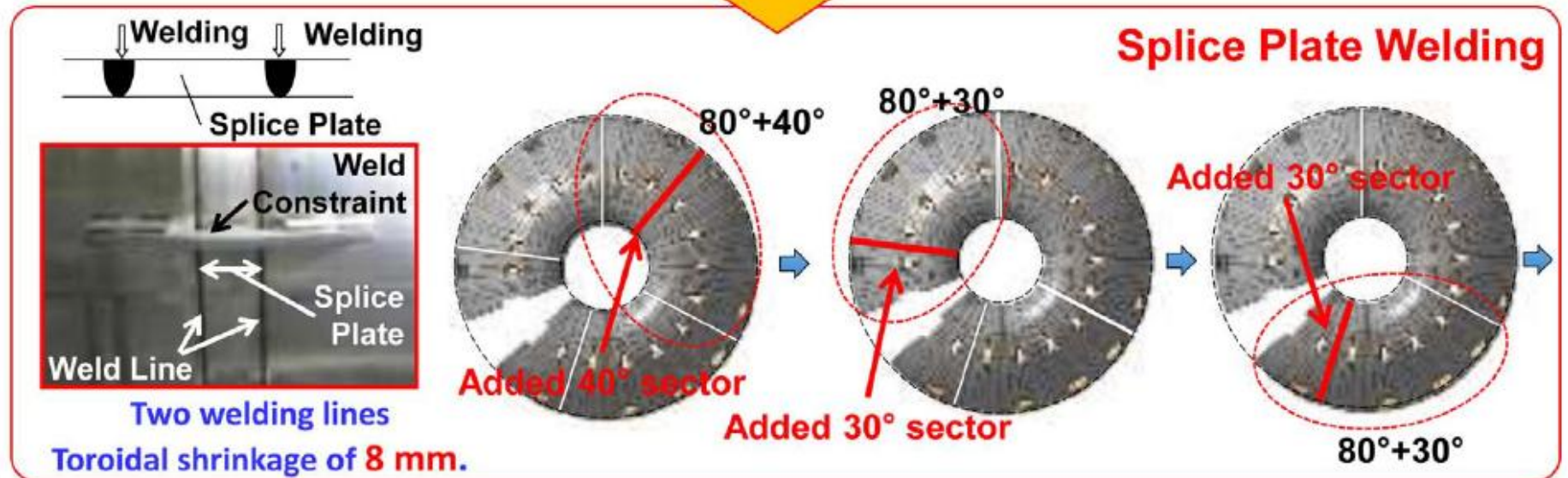
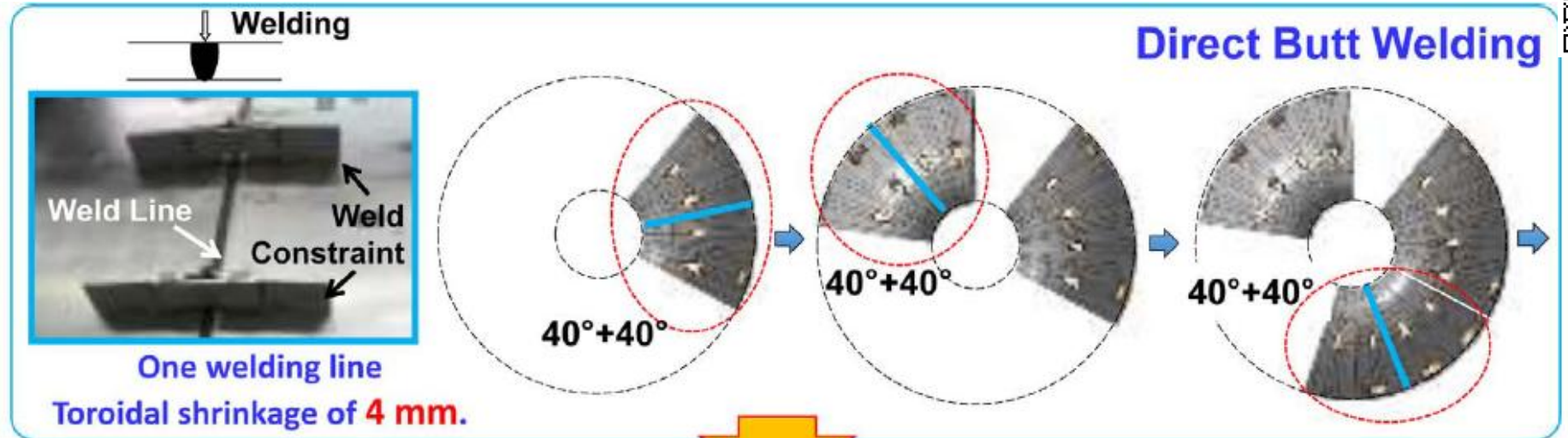
- Components of large experiments often have to withstand strong mechanical loads, therefore their mechanical connections have to be designed accordingly.
- Typically, removable connections are used for those components which might need to be replaced or maintained, and for those which have no pressure holding requirements.
- Bolted connections are commonly used for outer flanges, which need to be removed to connect new diagnostics, and for intercoil connections, where extensive welding might introduce excessive heat which might damage conductors and electrical insulation.
- Additionally, shear keys or shear pins might be needed to withstand large shear loads.



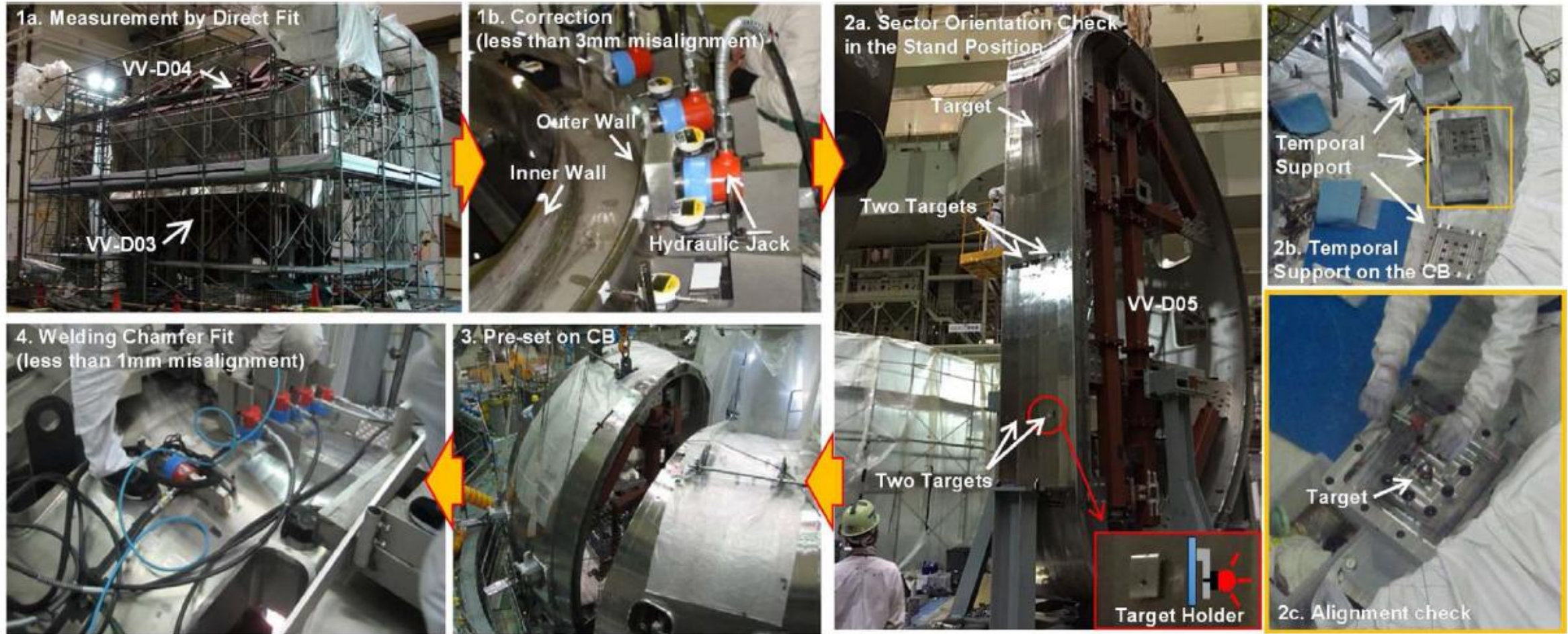
Large mechanical connections between the TF coils of JT-60SA

Pinned and bolted connections at the gravity support of JT-60SA TF coils

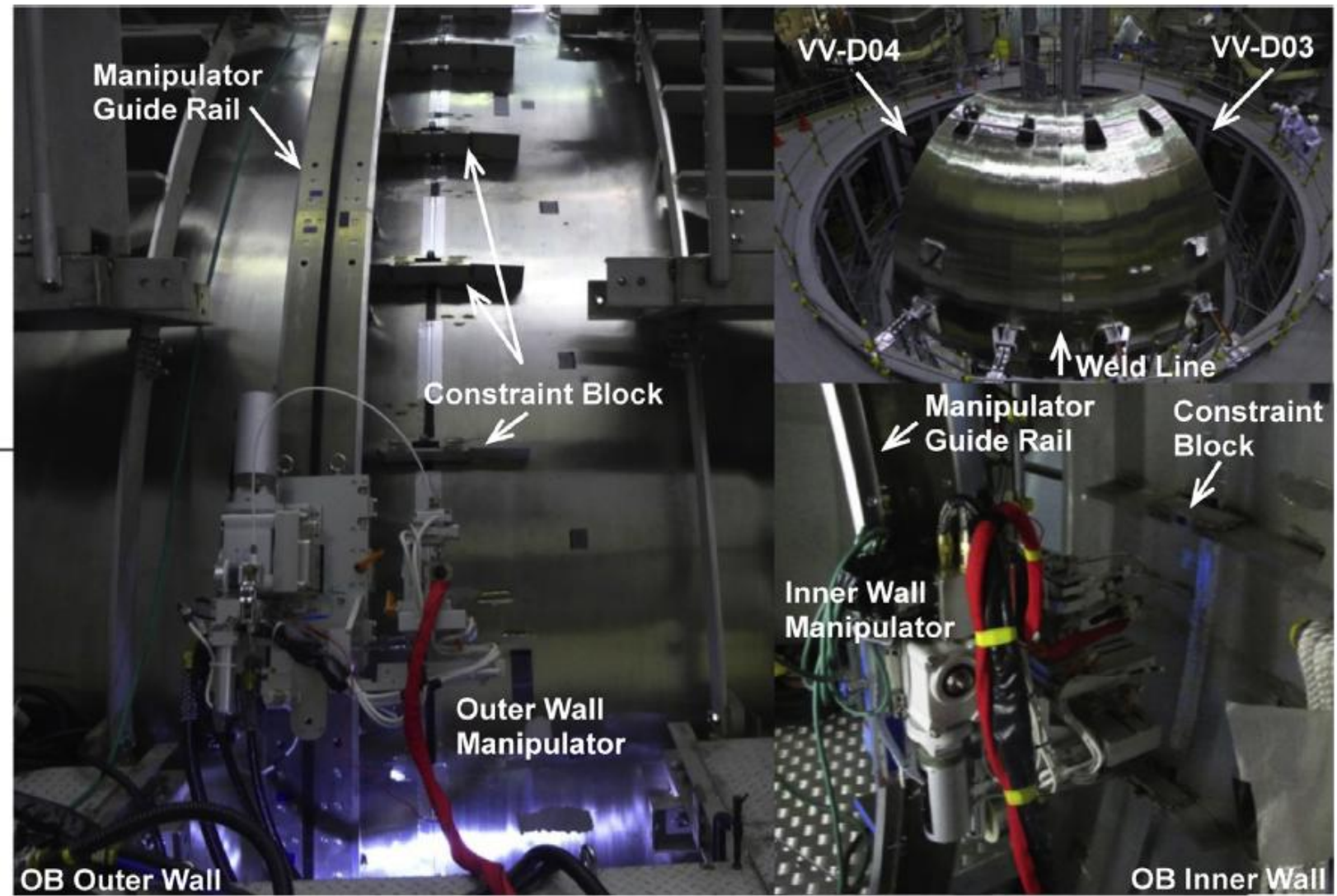
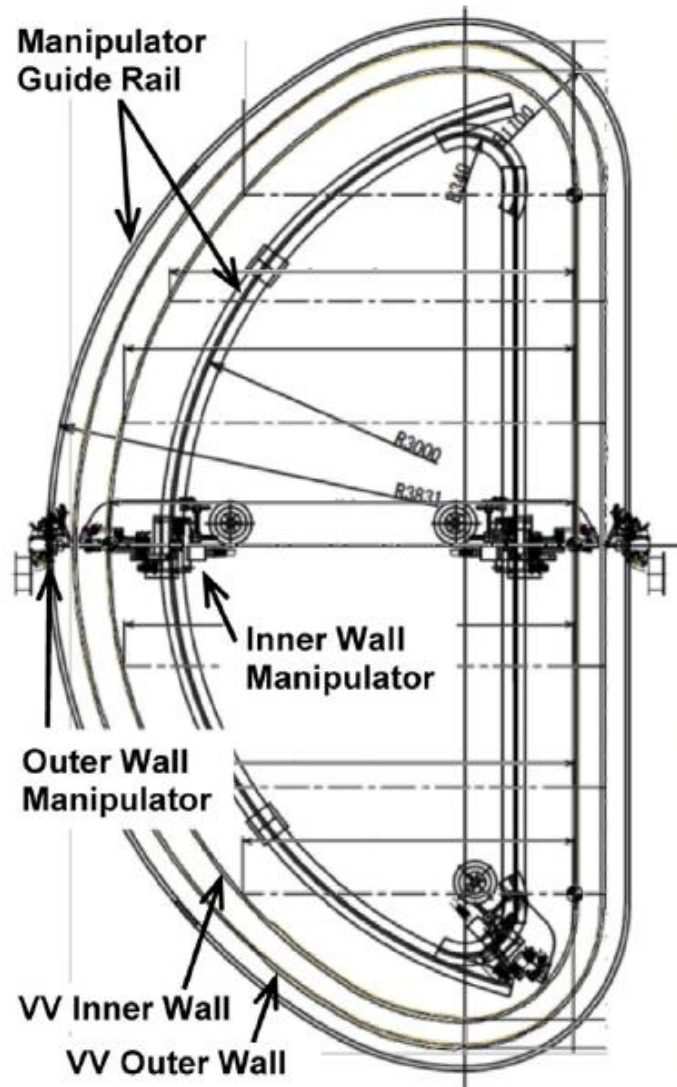
- Welded connections are preferred for all pressure boundaries, and for those components which are permanently connected together.
- The typical component which is always welded is the vacuum vessel, where the UHV compatibility is mandatory.
- The vacuum vessel welding is the most delicate of all welding operations on site, as the vacuum vessel is the main nuclear container of every nuclear fusion experiment, and its geometry is very challenging.
- In particular, the double walled design of the largest vacuum vessel, require careful planning and extensive validation of all welding procedures, and the nuclear nature of the component increases the quality standards to be used during NDT and QA.



Assembly sequence of the JT-60SA vacuum vessel



Fit-up of the vacuum vessel sectors



Welded connections of the JT-60SA vacuum vessel

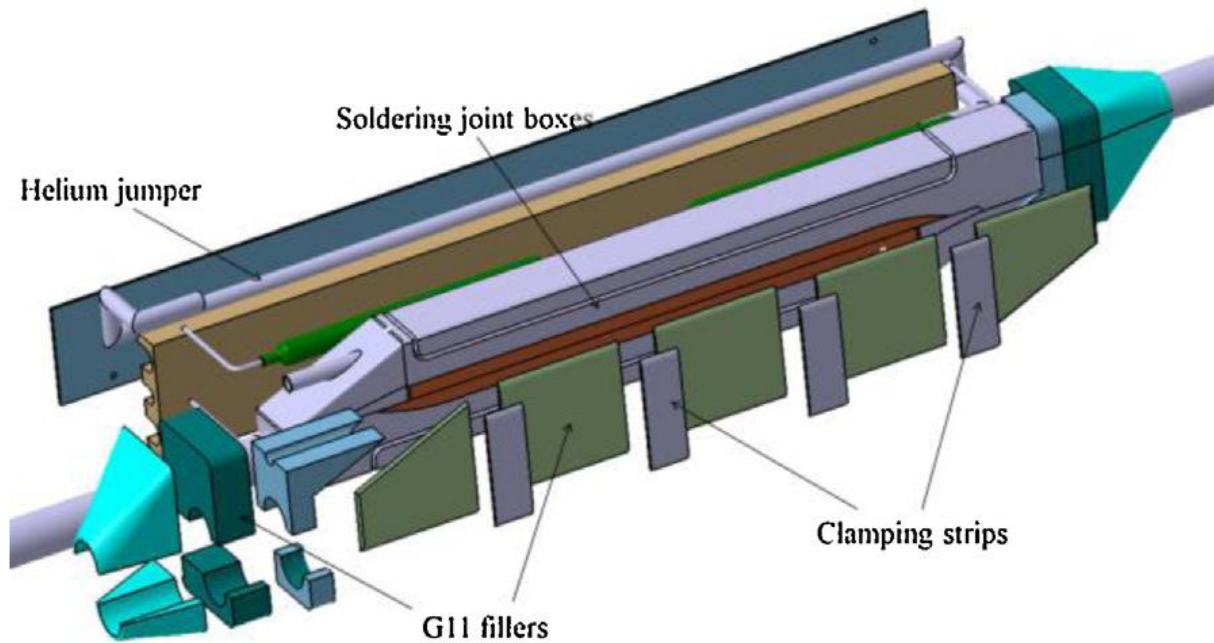
- For those components which require an electrical connection, like the magnet coils, special techniques must be developed to establish a reliable electrical connection, which can withstand many operational cycles (> 10000) without the possibility to be inspected or replaced.
- The electrical contact is normally obtained by either soldering, or by mechanical compression with an intermediate soft metal (e.g. Indium).
- After the electrical connection is prepared, the joint need to be insulated against ground. For a machine the size of ITER, the DC voltage against ground can be as high as 56kV.
- The perfect execution of these joints is vital for the success of any experiment. A failure in an HV insulation could escalate into an arc, which given the enormous energies at play (several GJ), could result in a devastating discharge able to melt metal and severely damage the component involved.



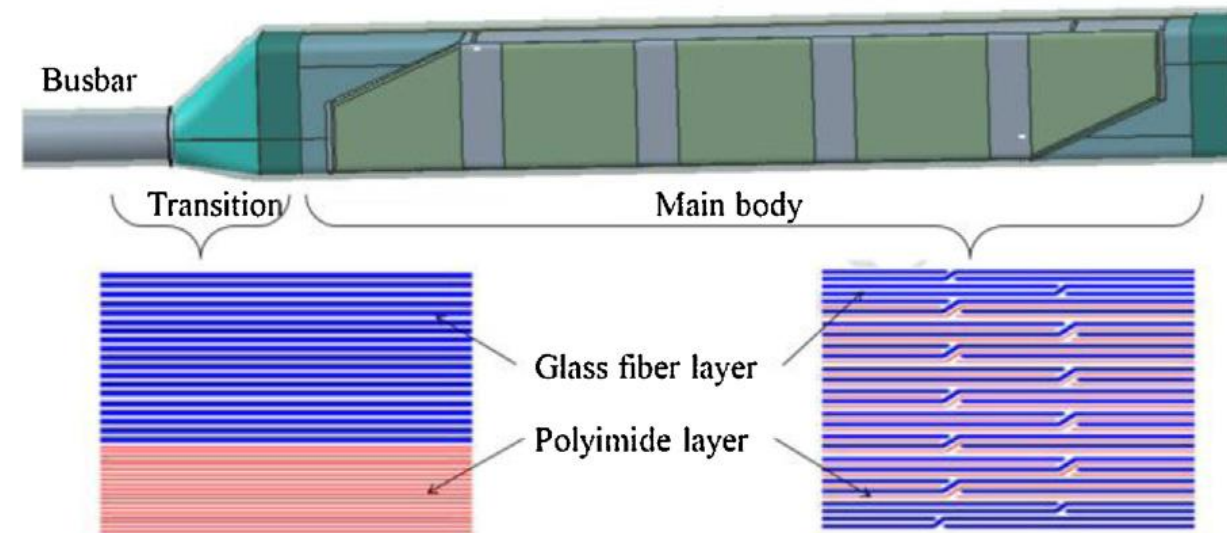
70 kA electrical feeder joint for the toroidal fields coils



A main busbar joint sample during qualification test



Exploded view of an ITER superconducting electrical joint



Insulation lay-up sequence



Application of glass-epoxy insulation by hand



Final component after VPI of glass-kapton-epoxy insulation

- Once a component is delivered, it undergoes incoming inspections, including visual test of the integrity of the packaging, checking of the acceleration sensors and temperature loggers, and also functional tests, like pressure, flow and leak tests.
- Careful dimensional inspection is of paramount importance for the smooth assembly of large components, optical metrology is used routinely for this purpose, by means of laser tracking, laser scanning and photogrammetry.
- Precise positioning of large components with tight tolerances is one of the most challenging activities during the installation of a large experiment. Dedicated tools must be engineered and manufactured to suit such demanding applications
- The final connection of large high technology components, both mechanical and electrical, is of great importance for the success of the experiment. Mechanical connections must be designed against operational and accidental loads and electrical connections must withstand as well operational and accidental voltages.